

The Cobordism Functor, the Theta Register, and Root Systems

Solutions

Solutions to the companion *Problems*, in its numbering; theorem and section numbers refer to `docs/design/cobordism_functor_reading.tex`. Where a problem asks for a value the verifier reports, the recorded value is quoted.

The primer

Cochains, the Laplacian and the harmonic space (Section 2)

P1. On a simplex $\sigma = [v_0 \dots v_{p+q+1}]$, with $(\alpha \cup \beta)[v_0 \dots v_{p+q}] = \alpha[v_0 \dots v_p] \beta[v_{p+1} \dots v_{p+q}]$,

$$d(\alpha \cup \beta)(\sigma) = \sum_{i=0}^{p+q+1} (-1)^i (\alpha \cup \beta)[v_0 \dots \hat{v}_i \dots v_{p+q+1}].$$

Split at $i \leq p$ and $i \geq p+1$. The first group is $(d\alpha \cup \beta)(\sigma)$ minus its $i = p+1$ term; the second is $(-1)^p(\alpha \cup d\beta)(\sigma)$ minus its $j = 0$ term. Those two omitted terms are $(-1)^{p+1}\alpha[v_0 \dots v_p]\beta[v_{p+1} \dots]$ and $(-1)^p\alpha[v_0 \dots v_p]\beta[v_{p+1} \dots]$, which cancel. Hence $d(\alpha \cup \beta) = d\alpha \cup \beta + (-1)^p\alpha \cup d\beta$. For closed α, β : $\langle \alpha|_{\partial W} \cup \beta|_{\partial W}, [\partial W] \rangle = \langle \alpha \cup \beta, \partial[W] \rangle = \langle d(\alpha \cup \beta), [W] \rangle = 0$.

P2. Every edge starts and ends at v , so $\partial_1 = 0$ and $d_0 = \partial_1^\top = 0$. With the ordering (a, b, c) and (U, L) , $\partial_2 = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $d_1 = \partial_2^\top = \begin{pmatrix} 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}$; $d_1 d_0 = 0$ trivially. $\ker d_1 = \{\alpha_c = \alpha_a + \alpha_b\}$ has dimension 2 and $\text{im } d_0 = 0$, so $b_1 = 2$; $b_0 = 1$ ($C^0 = \mathbb{C}$, $d_0 = 0$); $b_2 = 2 - \text{rank } d_1 = 1$. With $M_0 = 1$, $\delta_1 = M_0^{-1} d_0^\top M_1 = 0$, so $L_1 = \delta_2 d_1 = M_1^{-1} d_1^\top M_2 d_1$. For α with $x = \alpha_a + \alpha_b - \alpha_c$, $d_1 \alpha = (x, -x)$ and $d_1^\top M_2(x, -x)^\top = (m_U + m_L)x(1, 1, -1)^\top$. If $m_L = -m_U$ this vanishes for every α : $L_1 = 0$ and $\ker L_1 = C^1$, dimension 3. The closedness certificate fails: $d_1 \alpha = (x, -x) \neq 0$ for $\alpha \notin \ker d_1$ although $L_1 \alpha = 0$; coclosedness holds vacuously and the span test would report dimension 3 against $b_1 = 2$. Theorem 2.2 needs positive Hermitian masses, for which $\langle h, L_1 h \rangle_{M_1} = \|d_1 h\|_{M_2}^2 + \|\delta_1 h\|_{M_0}^2$ forces $d_1 h = 0$; with M_2 indefinite the “energy” $(m_U + m_L)|x|^2$ can vanish on a nonzero $d_1 h$. This is the Lorentzian caveat after Theorem 2.2.

P3. Two representatives ω, ω' of the same class α differ by $d\eta$; by Lemma 2.1, $\langle d\eta, z \rangle = \langle \eta, \partial z \rangle = 0$ for every cycle z , so the periods agree. J1 (fixed M) is the corollary to Theorem 4.2: M is built from periods only. J2 (the representative moves by an $O(1)$ amount) measures that the harmonic condition $\delta\omega = 0$ depends on the masses, so the chosen cochain in the class changes. J3 (the difference is exactly in $\text{im } d_0$) measures the proof of Theorem 2.4: if the difference had a non-exact part, the two cochains would lie in different classes, which cannot happen for a change of lengths on a fixed complex; it would signal a certificate failure (the band was not the harmonic space) or a singular P_A .

Restriction and the monodromy (Sections 3–4)

P4. Lefschetz gives $H^0(W, \partial W) \cong H_3(T^2) = 0$, $H^1(W, \partial W) \cong H_2(T^2) = \mathbb{Z}$, $H^2(W, \partial W) \cong H_1(T^2) = \mathbb{Z}^2$, $H^3(W, \partial W) \cong \mathbb{Z}$. With $H^0(W) = \mathbb{Z}$, $H^0(\partial W) = \mathbb{Z}^2$, $H^1(W) = \mathbb{Z}^2$, $H^1(\partial W) = \mathbb{Z}^4$, $H^2(W) = \mathbb{Z}$, $H^2(\partial W) = \mathbb{Z}^2$ the sequence is

$$0 \rightarrow 0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}^2 \xrightarrow{\iota^*} \mathbb{Z}^4 \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z} \rightarrow 0.$$

$H^0(W) \rightarrow H^0(\partial W)$ is the diagonal, rank 1, so its cokernel $\mathbb{Z}$ maps *onto* $H^1(W, \partial W) = \mathbb{Z}$, so $H^1(W, \partial W) \rightarrow H^1(W)$ is zero, so ι^* is injective: $\text{rank } \iota^* = 2 = \frac{1}{2} \cdot 4$. Consistency: the next map has rank $4 - 2 = 2$, onto $\mathbb{Z}^2$, so $H^2(W) \rightarrow H^2(\partial W)$ is injective of rank 1 and $H^2(\partial W) \rightarrow H^3(W, \partial W)$ has rank 1, onto.

P5. W retracts onto T^2 and both ends restrict by the identity, so $L_W = \{(p, p) : p \in \mathbb{C}^2\}$, the diagonal. For $\Omega_\partial = J \oplus (-J)$: $(p, p)^\top \Omega_\partial(q, q) = p^\top Jq - p^\top Jq = 0$. Dimension 2 in $\mathbb{C}^4$: Lagrangian.

P6. $J\sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, so $\sigma^\top J\sigma = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -J$. Hence $(\sigma \oplus \sigma)^\top (J \oplus J)(\sigma \oplus \sigma) = -(J \oplus J)$ while $\det(\sigma \oplus \sigma) = (-1)^2 = +1$. The 2×2 identity would predict $+(J \oplus J)$. Each block reverses the form independently; the reversal is a property of each torus's relabelling, so the sign must be declared per torus. (The code first tied it to $\det M$ and this counterexample corrected it.)

P7. The far marking is $(\varphi A, \varphi B)$, so $P_B = \varphi^* P_A$ and $M = \varphi^*$. Swap: exchanges A, B ; $\varphi^* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\det = -1$. Dehn twist $(x, y) \mapsto (x+y, y)$: $A \mapsto A, B \mapsto A+B$, so $\varphi^* = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\det = +1$. On the grid with diagonal $(i, j) - (i+1, j+1)$ the swap sends that diagonal to $(j, i) - (j+1, i+1)$, a diagonal of the same family: simplicial. The twist $(i, j) \mapsto (i+j, j)$ sends the diagonal to $(i+j, j) - (i+j+2, j+1)$, a displacement $(2, 1)$, which is not an edge: not simplicial. Automorphisms of a finite triangulation have finite order, and a twist has infinite order, so no automorphism realizes it. A flip pass changes the triangulation: flipping every edge of one family of the far torus produces the same grid drawn in a sheared basis, i.e. the image of the grid under the lattice map L . Each flip is a 2–2 Pachner move, which is the boundary of one tetrahedron; stacking one tetrahedron per flipped edge gives a layer whose near face is the old grid and whose far face is the sheared grid. The far marking is read through L , so the collar's relation is $M = L^\top$; the measured passes give $T_B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, T_A and S (F2).

P8. 1.6×10^{-14} measures the corollary to Theorem 4.2 (metric independence of M), whose mechanism is Theorem 2.4: periods of a fixed class do not move. The 0.7 move with an exact difference measures Theorem 2.4's proof: the two harmonic representatives of one class differ by a coboundary. A nonzero non-exact part would mean the representatives lay in *different* classes — impossible for a change of lengths on a fixed complex — so it would signal that the band was not the harmonic space (a certificate failure) or that P_A was singular and the canonical normalization failed.

P9. $M = P_B P_A^{-1} = \text{diag}(2, \frac{1}{2})$, $\det M = 1$, and for 2×2 matrices $\det = 1$ is symplecticity, but $\frac{1}{2} \notin \mathbb{Z}$. Theorem 4.2 assumes that the two restrictions identify the bulk lattice with each boundary lattice $H^1(M_i; \mathbb{Z})$; then M is the matrix of a $\mathbb{Z}$ -module isomorphism, hence integral with unit determinant. Invertibility over $\mathbb{C}$ says only that the restriction is injective on a basis, not that it hits the integral basis. Connected sum: by Mayer–Vietoris with $W_1 \cap W_2 \simeq S^2$ (Proposition 16.1), $H^1(W) = H^1(T^2 \times I) \oplus H^1(S^1 \times D^2) = \mathbb{Z}^2 \oplus \mathbb{Z}$, and the new class is supported away from the

boundary tori, so its periods on both ends vanish. With a harmonic basis of three forms, P_A and P_B are 2×3 with the invisible class in $\ker P_A \cap \ker P_B$. The identifiability proposition: $B = TA$ has a unique solution on $\text{im } A$ exactly when A is onto and $\ker A \subseteq \ker B$; both hold, so M is the unique map with $P_B = MP_A$, and it equals the product collar's I .

Choi vectors and composition (Sections 5–6)

P10. Let $\pi_A : L \rightarrow A$ be the projection. $\ker \pi_A = L \cap (0 \oplus B)$, so π_A is injective iff that is zero, and it is onto by hypothesis; then $\pi_A|_L$ is an isomorphism and $T := \pi_B \circ (\pi_A|_L)^{-1}$ is the unique linear map with $L = \{(x, Tx)\}$. Conversely a graph meets $0 \oplus B$ trivially and projects onto A . $\dim L = \dim A$.

P11. $|I\rangle\rangle = e_1 \otimes e_1^* + e_2 \otimes e_2^*$, the Bell vector, Schmidt rank two; $|\sigma\rangle\rangle = e_2 \otimes e_1^* + e_1 \otimes e_2^*$, also maximally entangled. The product collar's Choi vector is maximally entangled because the identity channel correlates input and output perfectly: under Choi, unitary channels correspond exactly to maximally entangled operator vectors. This is operator-vector information (Section 29), not entanglement of a physical two-torus state.

P12. With $U = W_1$, $V = W_2$, $U \cap V = T^2 \times \{1\}$: $H^0(U) \oplus H^0(V) \rightarrow H^0(T^2)$ is $(c_1, c_2) \mapsto c_1 - c_2$, onto, so $H^1(W) \rightarrow H^1(W_1) \oplus H^1(W_2)$ is injective with image the kernel of $(\alpha_1, \alpha_2) \mapsto \alpha_1|_{T^2} - \alpha_2|_{T^2}$, i.e. compatible pairs $\cong H^1(T^2) = \mathbb{Z}^2$. The far ends read the compatible pair's two restrictions, both the identity: L_W is the diagonal and $M = I \cdot I = I$. The result is the product collar $T^2 \times [0, 2]$.

P13. $(\subseteq)$ A class α on W restricts to α_1, α_2 on W_1, W_2 which agree on M_1 (both are $\alpha|_{M_1}$); then $(\alpha|_{M_0}, \alpha|_{M_1}) \in L_{W_1}$ and $(\alpha|_{M_1}, \alpha|_{M_2}) \in L_{W_2}$, so $(\alpha|_{M_0}, \alpha|_{M_2}) \in L_{W_2} \circ L_{W_1}$ with $x_1 = \alpha|_{M_1}$. $(\supseteq)$ Given $(x_0, x_1) \in L_{W_1}$ and $(x_1, x_2) \in L_{W_2}$, choose $\alpha_1 \in H^1(W_1)$ restricting to (x_0, x_1) and $\alpha_2 \in H^1(W_2)$ restricting to (x_1, x_2) . They agree on M_1 , so (α_1, α_2) lies in the kernel of $H^1(W_1) \oplus H^1(W_2) \rightarrow H^1(M_1)$, which by exactness is the image of $H^1(W)$: some α restricts to both, hence to (x_0, x_2) . Exactness at $H^1(W_1) \oplus H^1(W_2)$ is the step used. The connecting map $H^0(M_1) \rightarrow H^1(W)$ makes α non-unique (two extensions differ by an invisible class), but every extension has the same restrictions, so the relation is unaffected; only the harmonic space, not L_W , sees the extra class. On a simplicial complex the sequence requires one cell structure on the seam, which is why separately seeded collars do not compose by this theorem.

P14. The $2L$ -layer collar is seeded as one complex; nothing identifies interface cells of two L -layer collars, and no harmonic form is required to be continuous across a seam, so Mayer–Vietoris is never exercised. The check verifies $M = \varphi^*$ on two hosts and the relation $\sigma^2 = I$ in the group. The automorphisms realizable on the square grid, $\{\pm I, \pm \sigma\} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$, commute, so among automorphisms no experiment distinguishes $M(W_2 \circ W_1)$ from $M(W_1 \circ W_2)$. The flip passes stack two layers on one host, so the far marking is read through $L_2 L_1$ and the relation composes on a single complex: $T_A T_B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $T_B T_A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ differ (F4), and their Weil matrices are at projective distance 1 (T9). This is Theorem 6.1 on one complex with the seam being a face layer of the same triangulation. It still does not glue two separately seeded cobordisms along a shared boundary torus: that requires identifying interface cells and frames of two complexes.

What the metric controls (Section 7)

P15. Under the gauge the dressed metric transforms by $G \mapsto \rho G \rho^{-1}$ and the primal frame by $\Phi \mapsto \rho \Phi$. Then $\Phi^\top G \Phi \mapsto \Phi^\top \rho^\top \rho G \rho^{-1} \rho \Phi = \Phi^\top \rho^2 G \Phi$ (ρ diagonal), which differs from $\Phi^\top G \Phi$ unless $\rho^2 = I$. The dual frame transforms by $\tilde{\Phi} \mapsto \rho^{-1} \tilde{\Phi}$, so $\tilde{\Phi}^\top G \Phi \mapsto \tilde{\Phi}^\top \rho^{-1} \rho G \rho^{-1} \rho \Phi = \tilde{\Phi}^\top G \Phi$. At zero phases the dual connection equals the connection, $\Phi^\vee = \Phi$, and $\tilde{\Phi} = G \Phi^\vee B^{-\top} = Z B^{-\top}$ with $B = \Phi^\top Z$; then $\tilde{\Phi}^\top M Z = B^{-1} Z^\top M Z = B^{-1} B = I$ and the two Grams coincide. Measured: 0.76 against 1.3×10^{-15} ; agreement at zero phases to 3.6×10^{-16} .

P16. Every symbol in D is boundary data: G_A, G_B are the own-boundary Grams (functions of the boundary lengths), the markings are fixed, and $M, M^\vee$ are period transports, which by Theorem 2.4 depend only on the classes. No argument depends on interior lengths, so no bulk relaxation changes D ; a verifier that “expects the defect to fall” would be measuring nothing (this is why V2 keeps it as a control). $D = 0$ says the bilinear pairing G is preserved, $(M^\vee)^\top G_B M = G_A$, with G a complex bilinear form; unitarity is preservation of a positive Hermitian form, $T^\dagger Q_{\text{out}} T = Q_{\text{in}}$ (Section 29). A complex bilinear form is not an inner product, so the two statements are different tests.

Weights, phase space and quantization (Sections 8–9)

P17. A character $\chi : \mathfrak{h}/\Lambda \rightarrow U(1)$ lifts to a homomorphism $\mathfrak{h} \rightarrow \mathbb{R}$, $h \mapsto \langle \mu, h \rangle$, trivial on Λ iff $\langle \mu, \Lambda \rangle \subseteq \mathbb{Z}$, i.e. $\mu \in \Lambda^*$. On $V \otimes V'$, T acts diagonally, $t \cdot (v \otimes v') = \chi_\mu(t) \chi_{\mu'}(t) v \otimes v' = \chi_{\mu+\mu'}(t) v \otimes v'$: weights add. For $T_1 \times T_2$ a character is a pair of characters, so $\Lambda^* = \Lambda_1^* \oplus \Lambda_2^*$. The space spanned by all characters of $J(\Sigma) = H^1(\Sigma; \mathbb{R})/H^1(\Sigma; \mathbb{Z})$ is L^2 of a torus, infinite-dimensional, with no distinguished finite-dimensional subspace. Theta quantization chooses a complex structure (the period matrix Ω), a positive line bundle L whose curvature is the intersection form (the principal polarization) and a level k ; the holomorphic sections $H^0(J, L^k)$ are finite-dimensional (k^g) because holomorphy plus the quasi-periodicity fixes every Fourier coefficient from k^g residue classes (Theorem 11.2). A disjoint union gives a product Jacobian with the external product bundle, whose sections are the tensor product (Proposition 11.1).

P18. A gauge transformation adds $d\chi$ to ϕ , and $\oint_\gamma d\chi = 0$ around any cycle (Lemma 2.1), so a, b are unchanged. A large gauge transformation is a closed cochain with periods in $2\pi\mathbb{Z}$ (the exponential of i times it is a well-defined phase), so it shifts (a, b) by integers. Hence the physical configuration space is $H^1(\Sigma; \mathbb{R})$ modulo $H^1(\Sigma; \mathbb{Z})$, a torus of dimension $b_1 = 2g$ with coordinates $a_i, b_i \in \mathbb{R}/\mathbb{Z}$. On the basis dual to a symplectic marking the intersection form is $\begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ and the 2-form $\sum_i da_i \wedge db_i$ takes the value 1 on each (a_i, b_i) square and 0 on the others: its periods on H_2 of the torus are the entries of the intersection form, a unimodular integral class, the principal polarization. Shifting $b \mapsto b + e_i$ gives $z \mapsto z + e_i$; shifting $a \mapsto a + e_i$ gives $z \mapsto z - \Omega e_i$. Together these generate $\mathbb{Z}^g + \Omega \mathbb{Z}^g$, and the map $(a, b) \mapsto z = b - \Omega a$ is a real-linear isomorphism $\mathbb{R}^{2g} \rightarrow \mathbb{C}^g$ because $\text{Im } \Omega$ is invertible. So z descends to an isomorphism of $J(\Sigma)$ with the complex torus $\mathbb{C}^g/(\mathbb{Z}^g + \Omega \mathbb{Z}^g)$; the complex structure is the choice of Ω .

P19. $d_1 d_0 = -(v-1)(u-1) + (u-1)(v-1) = 0$. If $(u, v) \neq (1, 1)$ then $d_0 \neq 0$ has rank 1, so $H^0 = \ker d_0 = 0$; $d_1 \neq 0$ has rank 1, so $\ker d_1$ has dimension $1 = \text{rank } d_0$ and $H^1 = \ker d_1 / \text{im } d_0 = 0$; $H^2 = \mathbb{C} / \text{im } d_1 = 0$. The Euler characteristic $1 - 2 + 1 = 0$ is consistent. So inserting the holonomy phases into the differential (a twisted pencil) empties the register: there is no harmonic space left

to carry a state. The phases are configuration data (the point z of the Jacobian), read as Wilson loops of a configuration on the neutral complex; the Hodge assembly must stay untwisted.

Theta functions and the tensor register (Sections 10–11)

P20. $z \mapsto z + 1$ multiplies the n -th term by $e^{2\pi i k(n+j/k)} = e^{2\pi i(kn+j)} = 1$. $z \mapsto z + \tau$: the exponent gains $2\pi i k(n + j/k)\tau$; write $k(n + j/k)^2\tau + 2k(n + j/k)\tau = k[(n + 1 + j/k)^2 - 1]\tau$ and $2\pi i k(n + j/k)z = 2\pi i k(n + 1 + j/k)z - 2\pi i k z$; reindexing $n + 1 \rightarrow n$ gives $\theta_j(z + \tau) = e^{-\pi i k \tau - 2\pi i k z} \theta_j(z)$.

P21. Under $z \mapsto z + 1$ both factors are unchanged. Under $z \mapsto z + \tau$: $|\theta_j|$ is multiplied by $|e^{-\pi i k \tau - 2\pi i k z}| = e^{\pi k \operatorname{Im} \tau + 2\pi k \operatorname{Im} z}$, while $e^{-\pi k(\operatorname{Im} z + \operatorname{Im} \tau)^2 / \operatorname{Im} \tau} = e^{-\pi k(\operatorname{Im} z)^2 / \operatorname{Im} \tau} e^{-2\pi k \operatorname{Im} z} e^{-\pi k \operatorname{Im} \tau}$. The product of the two extra factors is 1.

P22. Shift the parallelogram so no zero lies on its boundary. The winding number of θ_j around it is $\frac{1}{2\pi i} \oint d \log \theta_j$. The two vertical sides cancel by periodicity. On the top side $d \log \theta_j(z + \tau) = d \log \theta_j(z) - 2\pi i k dz$, traversed backwards, so the horizontal sides contribute $\int_0^1 2\pi i k dz = 2\pi i k$: exactly k zeros. For $k = 2$: pairing $n \leftrightarrow -n - 1$ in the sum shows $\theta_1(\frac{1}{4}) = \theta_1(\frac{3}{4}) = 0$ (the paired terms carry $e^{\pi i(n + \frac{1}{2})}$ and its negative), and $\theta_0(\frac{1}{4} + \frac{\tau}{2}) = \theta_0(\frac{3}{4} + \frac{\tau}{2}) = 0$ (there the paired terms carry $(-1)^n$ and $(-1)^{-n-1}$).

P23. $\theta_j(z + \frac{1}{k})$: the term gains $e^{2\pi i(n+j/k)} = e^{2\pi i j/k}$, independent of n . $\theta_j(z + \frac{\tau}{k})$: the exponent gains $2\pi i(n + j/k)\tau$; with $k(n + j/k)^2 + 2(n + j/k) = k(n + \frac{j+1}{k})^2 - \frac{1}{k}$ and $2\pi i k(n + j/k)z = 2\pi i k(n + \frac{j+1}{k})z - 2\pi i z$ this is $e^{-\pi i \tau/k - 2\pi i z} \theta_{j+1}(z)$. Then $ZX\theta_j = e^{2\pi i(j+1)/k}(\dots)\theta_{j+1}$ and $XZ\theta_j = e^{2\pi i j/k}(\dots)\theta_{j+1}$ with the same prefactor, so $ZX = e^{2\pi i/k}XZ$.

P24. For $\Omega = \operatorname{diag}(\tau_1, \tau_2)$ the quadratic form is $\tau_1 m_1^2 + \tau_2 m_2^2$ and the linear term $m_1 z_1 + m_2 z_2$, so $\exp(\dots)$ is a product of a function of (n_1, z_1) and one of (n_2, z_2) and the double sum factorizes. An off-diagonal Ω_{12} contributes $2\pi i k m_1 m_2 \Omega_{12}$, which couples the indices.

P25. Write $\theta_j(x + \tau y) = \sum_n e^{\pi i k(n+j/k)^2 \tau + 2\pi i k(n+j/k)(x + \tau y)}$. The x -frequencies $k(n + j/k) = kn + j$ are distinct integers for distinct (n, j) , so $\int_0^1 \overline{\theta_j} \theta_{j'} dx$ vanishes unless $j = j'$ and pairs equal n : it equals $\sum_n |e^{\pi i k(n+j/k)^2 \tau + 2\pi i k(n+j/k)\tau y}|^2 = \sum_n e^{-2\pi k \operatorname{Im} \tau [(n+j/k)^2 + 2(n+j/k)y]}$. Multiplying by $e^{-2\pi k y^2 \operatorname{Im} \tau}$ completes the square: $\sum_n e^{-2\pi k \operatorname{Im} \tau (n+j/k+y)^2}$. Integrating y over $[0, 1]$ and summing over n tiles the real line once, giving $\int_{\mathbb{R}} e^{-2\pi k \operatorname{Im} \tau t^2} dt = (2k \operatorname{Im} \tau)^{-1/2}$, cancelled by the prefactor. The weight is the square of $e^{-\pi k(\operatorname{Im} z)^2 / \operatorname{Im} \tau}$ because $\operatorname{Im} z = y \operatorname{Im} \tau$. For general g , with $z = x + \Omega y$ and $Y = \operatorname{Im} \Omega$, the same steps give $\int_{\mathbb{R}^g} e^{-2\pi k t^T Y t} dt = (2k)^{-g/2} (\det Y)^{-1/2}$, hence the prefactor $(2k)^{g/2} \sqrt{\det Y}$.

The geometric operator algebra (Section 12)

P26. $Z_r X_s |j\rangle = Z_r |j + \epsilon_s\rangle = \omega^{j_r + \delta_{rs}} |j + \epsilon_s\rangle$ and $X_s Z_r |j\rangle = \omega^{j_r} |j + \epsilon_s\rangle$, so $Z_r X_s = \omega^{\delta_{rs}} X_s Z_r$. Iterating, $Z^q X^{p'} = \omega^{q \cdot p'} X^{p'} Z^q$. Then $D_{p,q} D_{p',q'} = X^p Z^q X^{p'} Z^{q'} = \omega^{q \cdot p'} X^{p+p'} Z^{q+q'}$. For the adjoint, $X^\dagger = X^{-1}$ and $Z^\dagger = Z^{-1}$, so $D_{p,q}^\dagger = Z^{-q} X^{-p} = \omega^{(-q) \cdot (-p)} X^{-p} Z^{-q} = \omega^{p \cdot q} D_{-p, -q}$. Associativity: $(D_{p,q} D_{p',q'}) D_{p'',q''} = \omega^{q \cdot p' + (q+q') \cdot p''} D_{p+p'+p'', \dots}$ and $D_{p,q} (D_{p',q'} D_{p'',q''}) = \omega^{q' \cdot p'' + q \cdot (p'+p'')} D_{\dots}$; both exponents equal $q \cdot p' + q \cdot p'' + q' \cdot p''$, the cocycle condition for the bilinear form $(p, q), (p', q') \mapsto q \cdot p'$.

P27. $D_{p,q}^\dagger D_{p',q'} = \omega^{p,q} D_{-p,-q} D_{p',q'} = \omega^{p,q-q'p'} D_{p'-p,q'-q}$. If $p' \neq p$ the operator $X^{p'-p} Z^{q'-q}$ maps $|j\rangle$ to a multiple of $|j+p'-p\rangle \neq |j\rangle$: zero diagonal, zero trace. If $p' = p$ and $q' \neq q$ the trace is $\sum_j \omega^{(q'-q) \cdot j}$, a sum of a nontrivial character of $(\mathbb{Z}/k)^g$, which vanishes. If both agree the trace is d . Hence the $D_{p,q}$ are orthogonal for $\langle A, B \rangle = \text{Tr}(A^\dagger B)$; there are $d^2 = \dim \text{End}(\mathcal{Q}_k)$ of them, so they are a basis and the coefficient of T on $D_{p,q}$ is $\text{Tr}(D_{p,q}^\dagger T)/d$. For $a = \sum a_{p,q} e_{p,q}$, $b = \sum b_{p',q'} e_{p',q'}$, the product law gives $ab = \sum_{p,q,p',q'} a_{p,q} b_{p',q'} \omega^{q \cdot p'} e_{p+p',q+q'}$; collecting $r = p+p'$, $s = q+q'$ gives the stated formula. Example ($k=2$, $\omega = -1$): $a \star b$ has the single term $r = (1), s = (1)$ from $p=1, q=0, p'=0, q'=1$ with phase $\omega^{0 \cdot 0} = 1$: $a \star b = e_{1,1} = XZ$. $b \star a$: $p=0, q=1, p'=1, q'=0$ with phase $\omega^{1 \cdot 1} = -1$: $b \star a = -e_{1,1} = -XZ$. Indeed $ZX = \omega XZ = -XZ$.

P28. $|0\rangle\langle 0| = \frac{1}{2}(I + Z)$, so $a = \frac{1}{2}e_{0,0} + \frac{1}{2}e_{0,1}$ and $\text{tr}_G(a) = 2 \cdot \frac{1}{2} = 1$. $|+\rangle\langle +| = \frac{1}{2}(I + X)$, so $e = \frac{1}{2}e_{0,0} + \frac{1}{2}e_{1,0}$. By the twisted convolution, $a \star e = \frac{1}{4}e_{0,0} + \frac{1}{4}e_{1,0} + \frac{1}{4}e_{0,1} + \frac{1}{4}\omega^{1 \cdot 1}e_{1,1}$, whose $e_{0,0}$ coefficient is $\frac{1}{4}$; $\text{tr}_G(a \star e) = 2 \cdot \frac{1}{4} = \frac{1}{2}$, the probability of the outcome $+$ on $|0\rangle$. Positivity of a : $a = b^\star b$ with $b = a$ since $a \star a = a$ (a projector; check: the $e_{0,0}$ and $e_{0,1}$ coefficients of $a \star a$ are $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ each, using $e_{0,1} \star e_{0,1} = e_{0,0}$).

P29. In the basis $|j_1\rangle \otimes |j_2\rangle$ of Proposition 11.1, X_r and Z_r act on the factor containing r and trivially on the other, so $X^{(p_1,p_2)} Z^{(q_1,q_2)} = (X^{p_1} Z^{q_1}) \otimes (X^{p_2} Z^{q_2})$. The vectorization inner product is $\langle\langle A|B \rangle\rangle = \text{Tr}(A^\dagger B)$, so P27 gives orthonormality of $|D_{p,q}\rangle\rangle/\sqrt{d}$. With $|T\rangle\rangle = \sum_{ij} T_{ij} |i\rangle_{\text{out}} \otimes |j\rangle_{\text{ref}}$, $|T_2\rangle\rangle \otimes |T_1\rangle\rangle = \sum T_{2,ij} T_{1,kl} |i\rangle|j\rangle|k\rangle|l\rangle$; contracting the reference factor of T_2 against the output factor of T_1 (pairing $\langle j|k\rangle$) leaves $\sum_{il} (\sum_j T_{2,ij} T_{1,jl}) |i\rangle|l\rangle = |T_2 T_1\rangle\rangle$. For matrix units, $E_{ij} E_{kl} = \delta_{jk} E_{il}$ is exactly the contraction. The identity holds for unnormalized operator vectors: $\|T_2 T_1\|_F$ is not $\|T_2\|_F \|T_1\|_F$ in general, so renormalizing each factor to a unit state and then contracting would produce a vector whose norm no longer records the success amplitude of the composite (for a projector followed by a projector, the surviving weight).

The modular action and the Weil representation (Section 13)

P30. $\pi i k(n + j/k)^2(\tau + 1)$ adds $\pi i(kn^2 + 2nj + j^2/k)$; $e^{2\pi i n j} = 1$ always, $e^{\pi i k n^2} = (-1)^{kn^2} = 1$ for k even, leaving $e^{\pi i j^2/k}$. For k odd, $e^{\pi i k n^2} = (-1)^n$ depends on n , so $\theta_j(z; \tau + 1)$ is not a multiple of $\theta_j(z; \tau)$ (numerically the ratios $|\theta_j(\tau + 1)/\theta_j(\tau)|$ are not even constant in j); it is a theta function with characteristic shifted by $\frac{1}{2}$, and Θ_k is invariant only under a finite-index subgroup (the theta group generated by S and T^2).

P31. Write $\theta_j(z/\tau; -1/\tau) = \sum_n f(n + j/k)$ with $f(x) = e^{-\pi i k x^2/\tau + 2\pi i k x z/\tau}$. Poisson: $\sum_n f(n + a) = \sum_m \hat{f}(m) e^{2\pi i m a}$. The Gaussian integral $\int e^{-\pi a x^2 + 2\pi i b x} dx = a^{-1/2} e^{-\pi b^2/a}$ with $a = ik/\tau$, $b = kz/\tau - m$ gives $\hat{f}(m) = (\tau/ik)^{1/2} e^{\pi i k z^2/\tau} e^{-2\pi i z m} e^{\pi i m^2 \tau/k}$. Write $m = kn' + l$, $l \in \mathbb{Z}/k$: $m^2/k = k(n' + l/k)^2$ and $zm = k(n' + l/k)z$, so

$$\theta_j(z/\tau; -1/\tau) = (\tau/ik)^{1/2} e^{\pi i k z^2/\tau} \sum_l e^{2\pi i j l/k} \theta_l(-z; \tau) = (-i\tau)^{1/2} e^{\pi i k z^2/\tau} \sum_l \frac{e^{-2\pi i j l/k}}{\sqrt{k}} \theta_l(z; \tau),$$

using $\theta_l(-z) = \theta_{-l}(z)$ and $(\tau/ik)^{1/2} = (-i\tau)^{1/2} k^{-1/2}$ (principal branches, $\text{Im } \tau > 0$). So $\rho_k(S)_{jl} = k^{-1/2} e^{-2\pi i j l/k}$ exactly, with automorphy factor $(-i\tau)^{1/2}$ and no extra constant; at $k=2$, $e^{-\pi i j l} = (-1)^{jl}$ gives the Hadamard matrix, which is real, so the conjugate sign is invisible there. At $k=4$ the entries $e^{\mp \pi i j l/2}$ differ, and a numerical evaluation of both sides at one (z, τ) shows the residual 3×10^{-16} for $e^{-2\pi i j l/k}$ against 1.5 for the conjugate: the sign is forced.

P32. $(\rho(S)^2)_{jm} = \frac{1}{k} \sum_l e^{-2\pi i(j+m)l/k} = \delta_{j,-m}$: parity. For $k = 2$, $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $T = \text{diag}(1, i)$: $HT = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$, $(HT)^2 = \frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ -1 & i \end{pmatrix}$, $(HT)^3 = \frac{1+i}{\sqrt{2}} I = e^{i\pi/4} I$, while $S^2 = \text{parity} = I$ at $k = 2$. So $(ST)^3 = e^{i\pi/4} S^2$: the relation holds projectively, as the fit's group-law check reports (9.5×10^{-16}). (With the $c/24$ normalization of the affine T , P64, the relation becomes exact.)

P33. $(n + j/k)^\top B(n + j/k) = n^\top Bn + \frac{2}{k} n^\top Bj + \frac{1}{k^2} j^\top Bj$. Times πik : $\pi ik n^\top Bn = \pi ik \sum_i B_{ii} n_i^2 + 2\pi ik \sum_{i < l} B_{il} n_i n_l$, every term in $2\pi i\mathbb{Z}$ for k even; $2\pi i n^\top Bj \in 2\pi i\mathbb{Z}$; the remainder is $e^{\pi i j^\top B j / k}$. For $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $j^\top B j = 2j_1 j_2$, so the phase is $e^{2\pi i j_1 j_2 / k} = (-1)^{j_1 j_2}$ at $k = 2$: $\text{diag}(1, 1, 1, -1)$, controlled- Z .

P34. For block-diagonal M the Siegel action, the map on z and the automorphy factor all split, and by P24 the basis splits, so the transformation law is the tensor product of the two laws and $\rho(M_1 \oplus M_2) = \rho(M_1) \otimes \rho(M_2)$. $CZ = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Z$ is a sum of two products and not one: operator Schmidt rank 2.

P35. For fixed τ the functions $\theta_l(\cdot; \tau)$ are linearly independent (distinct clock eigenvalues, P23), so the coefficients ρ_{jl} in the law are uniquely determined for each τ . The derivations of P30–P31 produce them as roots of unity over $\sqrt{k}$ with no τ anywhere, so the unique coefficients are the same at every τ . Equivalently ρ_k is a representation of a finite group built from the finite Heisenberg group, which has no continuous parameter. Measured: 9×10^{-16} .

P36. Let $\omega_1, \dots, \omega_g$ be the holomorphic forms with $\oint_{A_i} \omega_j = \delta_{ij}$, $\oint_{B_i} \omega_j = \Omega_{ij}$. A period map acting on periods by $a' = Pa + Qb$, $b' = Ra + Sb$ corresponds to the new cycles $A' = PA + QB$, $B' = RA + SB$ (as chains), so $\oint_{A'_i} \omega_j = (P + Q\Omega)_{ij}$ and $\oint_{B'_i} \omega_j = (R + S\Omega)_{ij}$. Renormalizing $\omega' = \omega(P + Q\Omega)^{-1}$ makes the A' -periods the identity and the B' -periods $\Omega' = (R + S\Omega)(P + Q\Omega)^{-1}$. Matching with $(A\Omega + B)(C\Omega + D)^{-1}$ gives $A = S$, $B = R$, $C = Q$, $D = P$. For z : $z' = Ra + Sb - \Omega'(Pa + Qb) = (S - \Omega'Q)b + (R - \Omega'P)a$, and from $\Omega'(P + Q\Omega) = R + S\Omega$, $R - \Omega'P = -(S - \Omega'Q)\Omega$, so $z' = (S - \Omega'Q)(b - \Omega a)$. The standard identity $(C\Omega + D)^{-\top} = A - \Omega'C$ for symplectic M (equivalently M_{per} symplectic) reads $S - \Omega'Q = (P + Q\Omega)^{-\top}$. Genus one: $\tau' = (R + S\tau)/(P + Q\tau) = (M_{21} + M_{22}\tau)/(M_{11} + M_{12}\tau)$, which is the relation F3 measures between the flipped far torus's own modulus in the relabelled marking and the near modulus, to 8×10^{-16} . Feeding M_{per} itself into the Siegel formula would use $(P\tau + Q)/(R\tau + S)$, which is wrong.

P37. The Z_r commute and are diagonal in the theta basis; $|0\rangle$ is a joint eigenvector with all eigenvalues 1. Its orbit $\{X^p|0\rangle = |p\rangle\}$ runs over the basis, and Z^q acts on $|p\rangle$ by the character ω^{qp} , so the d vectors carry the d distinct joint eigencharacters: any invariant subspace containing one of them contains its whole shift orbit, hence everything, so the representation is irreducible. A compatible scalar lift of a symplectic relabelling sends the generators to new operators $X'^p Z'^q$ obeying the same relations with the same central phases, so it defines a second irreducible representation on the same space. Both are irreducible representations of the finite Heisenberg group with the same central character, so (Stone–von Neumann for finite groups, or directly: build the new joint eigenvector and map orbits to orbits) there is an intertwiner U with $U D_{p,q} U^{-1} = D'_{p,q}$. Since the $D_{p,q}$ span $\text{End}(\mathcal{Q}_k)$, any two intertwiners differ by an operator commuting with everything, a scalar. If the lift preserves adjoints, then $U^\dagger U$ commutes with every $D_{p,q}$, so it is a positive scalar; rescaling makes U unitary, unique up to a phase. A composite of compatible lifts has an intertwiner that is a composite of intertwiners, unique up to phase, so the lifts compose projectively.

Orientation reversal and coherent states (Sections 14–15)

P38. $\overline{\theta_j(z; \Omega)} = \sum_n \exp(-\pi i k(n + j/k)^\top \bar{\Omega}(n + j/k) - 2\pi i k(n + j/k)^\top \bar{z}) = \sum_n \exp(\pi i k(n + j/k)^\top (-\bar{\Omega})(n + j/k) + 2\pi i k(n + j/k)^\top (-\bar{z})) = \theta_j(-\bar{z}; -\bar{\Omega})$. $R_{\text{per}} : (a, b) \mapsto (a, -b)$ sends $z = b - \Omega a$ to $-b - \Omega a$, which is $-\bar{z}$ for the polarization $-\bar{\Omega}$; so R_{per} acts on the value column by complex conjugation K . $\sigma^\top J \sigma = -J$ (P6), and $N = \sigma R_{\text{per}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = S$ satisfies $N^\top J N = J$. Its Siegel matrix, by P36 with $P = 0, Q = -1, R = 1, S = 0$, is $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = S^{-1}$. Hence the swap acts on values by $\rho_2(S^{-1})K = HK$ up to a phase ($H^{-1} = H$): Hadamard composed with conjugation, antiunitary.

P39. Conjugating the law: $\overline{F_{\Omega'}(z')} = \bar{f} \bar{\rho} \overline{F_{\Omega}(z)}$. Norms: $\|\overline{F_{\Omega'}(z')}\| = |f| \|\bar{\rho} \overline{F_{\Omega}(z)}\| = |f| \|F_{\Omega}(z)\|$ because $\bar{\rho}$ is unitary when ρ is. Dividing, $c_{\Omega'}(z') = (\bar{f}/|f|) \bar{\rho} c_{\Omega}(z)$. The coherent state is the Riesz representative of evaluation, so it transforms by the conjugate representation: applying ρ instead of $\bar{\rho}$ gives a vector that is not the coherent state at the transported point (for the Hadamard, $\bar{\rho} = \rho$, which hides the error at level two for S , but $\bar{\rho}(T) = \text{diag}(1, -i) \neq \rho(T)$). Both ρ and $\bar{\rho}$ are faithful unitary representations of the same finite group, and their images span the same matrix algebra (the complex span of the translations is $\text{End}(\mathcal{Q}_k)$ in either), so no completeness is lost by the choice; only consistency between fitted and applied representations matters.

P40. On a genus-one curve, Riemann–Roch gives $h^0(D) - h^0(K - D) = \deg D$ and $K = 0$, so $h^0(D) = \deg D$ for $\deg D > 0$. Thus $h^0(L^2) = 2$ and, for any point p , $h^0(L^2(-p)) = 1$: the sections vanishing at p form a line, so both basis sections cannot vanish at p simultaneously (else $h^0(L^2(-p)) = 2$). Hence $[\theta_0 : \theta_1]$ is a well-defined holomorphic map $E \rightarrow \mathbb{CP}^1$. It is nonconstant (the two sections are independent), so by the open mapping theorem its image is open, and by compactness closed: all of $\mathbb{CP}^1$. Its degree is the number of zeros of a generic section $\alpha\theta_0 + \beta\theta_1$, which is $\deg L^2 = 2$ (P22); so every ray has two preimages generically. Composing with the antiholomorphic bijection $[u : v] \mapsto [\bar{u} : \bar{v}]$ of $\mathbb{CP}^1$ preserves surjectivity, so $z \mapsto c_\tau(z)$ is onto. For a qutrit the image of E in $\mathbb{CP}^2$ is a curve (real dimension 2) inside a manifold of real dimension 4: a proper subset. In general the coherent family has complex dimension at most g against $k^g - 1$.

P41. By Proposition 11.1, $\theta_{(j_1, j_2)}(z; \Omega) = \theta_{j_1}(z_1; \tau_1) \theta_{j_2}(z_2; \tau_2)$, so the value column is the Kronecker product $F_{\tau_1}(z_1) \otimes F_{\tau_2}(z_2)$; conjugating and normalizing preserves the product structure, and a product vector has one Schmidt coefficient, entropy 0 (measured 5×10^{-30}). With $\Omega_{12} \neq 0$ the lattice sum does not factor (P24), so the 2×2 coefficient matrix generically has rank two. The register’s tensor labels are a declared identification of $\mathcal{Q}_2(\Omega)$ with $\mathcal{Q}_2 \otimes \mathcal{Q}_2$ (through the identity monodromy); the entanglement is a property of the encoded state at the holonomy point, which depends on the polarization Ω , not of the operator $\rho_2(I) = I$.

Seams, factorization and the Clifford limit (Sections 16–17)

P42. $U = W_1, V = W_2, U \cap V \simeq S^2$: $H^0(W_1) \oplus H^0(W_2) \rightarrow H^0(S^2)$ is onto, and $H^1(S^2) = 0$, so $H^1(W) \rightarrow H^1(W_1) \oplus H^1(W_2)$ is an isomorphism. Every harmonic form is a pair with one member on each collar, so the restriction to the four tori is $M_{01} \oplus M_{23}$ and $\rho = \rho(M_{01}) \otimes \rho(M_{23})$ (Proposition 13.3). The off-block norm tests that the isomorphism commutes with restriction to the boundary: no harmonic form has periods on both collars’ tori.

P43. $U \cap V \simeq S^1$, so $H^1(W) \rightarrow H^1(W_1) \oplus H^1(W_2) \rightarrow H^1(S^1) = \mathbb{Z}$ with kernel $\{(\alpha_1, \alpha_2) : \alpha_1(a_1) = \alpha_2(a_2)\}$; $H^0(U) \oplus H^0(V) \rightarrow H^0(S^1)$ is onto, so nothing new enters from degree zero and $H^1(W)$

is that kernel, of rank 3. On the boundary the restriction subspace obeys $p_1(a_1) = p_2(a_2)$: the two tori's periods are coupled. The simplest symplectic matrix with a coupled graph is the shear $\begin{pmatrix} I & B \\ 0 & I \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, whose ρ_2 is controlled-Z (P33) of Schmidt rank 2 (P34).

P44. Cut the handle in its middle disk: $U \simeq W_1$, $V \simeq W_2$, $U \cap V \simeq D^2$, contractible. $H^0(U) \oplus H^0(V) \rightarrow H^0(D^2)$ is onto and $H^1(D^2) = 0$, so $H^1(W) = H^1(W_1) \oplus H^1(W_2)$ exactly as for the sphere: a 1-handle between two components is a path, $W_1 \cup \text{handle} \cup W_2 \simeq W_1 \vee W_2$. An annulus has $H^1(S^1) = \mathbb{Z}$, which is what produces the coupling of P43. The tube host is a 1-handle, so its relation is block-diagonal: this is topology, and G2 confirms it ($M = I_4$, off-block norm 2.6×10^{-15}). The far boundary, however, is one connected genus-two surface, whose period matrix is defined by its harmonic 1-forms for the boundary metric; the neck lets a harmonic form with periods on one torus leak into the other, so $\Omega_{12} \neq 0$ with a size set by the neck's waist and length. That is a metric statement about the polarization Ω of one surface, not about the cobordism's relation: as the neck pinches, $\Omega_{12} \rightarrow 0$ and the diagonal approaches the two tori's moduli (G5, the separating degeneration).

P45. $|++\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ and CZ flips the sign of $|11\rangle$: $CZ|++\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$. Its coefficient matrix $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is $\frac{1}{\sqrt{2}}H$, with two equal singular values $\frac{1}{\sqrt{2}}$: the reduced state is $I/2$, one bit of entanglement. $\text{SWAP}(|a\rangle \otimes |b\rangle) = |b\rangle \otimes |a\rangle$ is a product for every product input, yet $\text{SWAP} = \frac{1}{2} \sum_{P \in \{I, X, Y, Z\}} P \otimes P$ has operator Schmidt rank four. The witness: an operator is entangling if some product state is mapped to a state with nonzero reduced entropy; E5 evaluates the geometric gate on all 36 product stabilizer states and finds 16 of them mapped to Bell pairs (exactly one bit). Operator Schmidt rank measures operator-vector correlations (the Choi state), which SWAP has maximally while entangling nothing; Proposition 13.3 only shows that rank one is *not* entangling.

P46. S sends $\tau \mapsto -1/\tau$ and $z \mapsto z/\tau$, exchanging the roles of the lattice directions 1 and τ ; the translation by τ/k (X) becomes the translation by $1/k$ (Z) up to the automorphy phases, so $\rho(S)X\rho(S)^{-1} \propto Z$. Translations by k -torsion points form the Heisenberg–Weyl group, and a unitary conjugating it to itself lies in its normalizer, the Clifford group; for $k = 2$ the group generated by H and $\text{diag}(1, i)$ is the whole single-qubit Clifford group (order 24 modulo phases). Gottesman–Knill: a Clifford circuit acting on a stabilizer state can be tracked by updating $2n$ generators of its stabilizer, polynomial in n , so the theta register alone yields nothing beyond classical simulation. This does not contradict Theorem 12.2: the complex *span* of the translations is the full matrix algebra, while the *group* generated by the modular gates is finite.

P47. $\text{Sp}(2, \mathbb{F}_2) = \text{SL}(2, \mathbb{F}_2) \cong S_3$, so the local Cliffords reduce into $S_3 \times S_3$, and with the swap into $S_3 \wr \mathbb{Z}/2$, of order $6 \cdot 6 \cdot 2 = 72$. An element of order five in S_6 is a 5-cycle; $5 \nmid 72$, so no conjugate of $S_3 \wr \mathbb{Z}/2$ contains it. A local Clifford $A \otimes B$ (or $\text{SWAP} \cdot A \otimes B$) has symplectic image in that subgroup, so an element of order five modulo 2 cannot be one: its entangling capacity is forced by its order. For M : expanding along the third row, $\det M = 1$; $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ and a direct multiplication gives $M^T J M = J$; $M^2 = \begin{pmatrix} 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}$ and continuing, $M^5 = I$ (the characteristic polynomial is $x^4 + x^3 + x^2 + x + 1$, the fifth cyclotomic polynomial). Modulo 2, $M \not\equiv I$ and $M^5 \equiv I$, so the order is five. The measured gate has operator Schmidt rank four and sends 16 of 36 product stabilizer states to Bell pairs (E4, E5), as the argument predicts.

From a Lie algebra to its root system; the Weyl group (Sections 18–19)

P48. Let v be a highest-weight vector: $ev = 0$, $hv = \lambda v$. From $[h, f] = -2f$, $h(f^n v) = (\lambda - 2n)f^n v$. From $[e, f] = h$ and induction, $ef^n v = n(\lambda - n + 1)f^{n-1}v$: indeed $ef^n v = fef^{n-1}v + hf^{n-1}v = [(n-1)(\lambda - n + 2) + (\lambda - 2n + 2)]f^{n-1}v = n(\lambda - n + 1)f^{n-1}v$. In finite dimension the string terminates: let N be least with $f^{N+1}v = 0$, $f^N v \neq 0$; then $0 = ef^{N+1}v = (N+1)(\lambda - N)f^N v$ forces $\lambda = N$. The span of $v, f^1 v, \dots, f^N v$ is invariant under e, f, h , so by irreducibility it is the module: eigenvalues $\lambda, \lambda - 2, \dots, -\lambda$, each once, dimension $\lambda + 1$. Identification: $[h, e] = 2e$ says $\alpha(h) = 2$; the coroot is the element $\alpha^\vee = h$ of $\mathfrak{h}$ with $\langle \mu, \alpha^\vee \rangle = \mu(h)$; $\omega(h) = 1$ so $\alpha = 2\omega$; $P = \mathbb{Z}\omega$ (the h -eigenvalues are integers), $Q = \mathbb{Z}\alpha = 2\mathbb{Z}\omega$, $P/Q = \mathbb{Z}/2$; $s_\alpha(\omega) = \omega - \langle \omega, \alpha^\vee \rangle \alpha = \omega - 2\omega = -\omega$, so $W = \{\pm 1\}$. $|\omega|^2 = |\alpha|^2/4 = \frac{1}{2}$.

P49. $[\text{diag}(a), E_{ij}] = (a_i - a_j)E_{ij}$, so the roots are $\varepsilon_i - \varepsilon_j$ with root vectors E_{ij} . With $\langle \varepsilon_i, \varepsilon_j \rangle = \delta_{ij} - \frac{1}{3}$: $|\varepsilon_i - \varepsilon_j|^2 = 2$ and $\langle \alpha_1, \alpha_2 \rangle = \langle \varepsilon_1, \varepsilon_2 \rangle - \langle \varepsilon_1, \varepsilon_3 \rangle - \langle \varepsilon_2, \varepsilon_2 \rangle + \langle \varepsilon_2, \varepsilon_3 \rangle = -\frac{1}{3} + \frac{1}{3} - \frac{2}{3} - \frac{1}{3} = -1$. Cartan matrix $A_{ij} = 2\langle \alpha_i, \alpha_j \rangle / |\alpha_j|^2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$; $\cos \angle(\alpha_1, \alpha_2) = -1/2$, so 120° . The positive roots are $\alpha_1, \alpha_2, \alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3 = \theta$. Fundamental weights: $\omega_1 = a\alpha_1 + b\alpha_2$ with $\langle \omega_1, \alpha_1 \rangle = 2a - b = 1$, $\langle \omega_1, \alpha_2 \rangle = -a + 2b = 0$ gives $\omega_1 = \frac{1}{3}(2\alpha_1 + \alpha_2)$, and by symmetry $\omega_2 = \frac{1}{3}(\alpha_1 + 2\alpha_2)$; $P/Q \cong \mathbb{Z}/3$ generated by ω_1 ($3\omega_1 \in Q$, $2\omega_1 = \omega_2 + \alpha_1$). $\rho = \omega_1 + \omega_2 = \alpha_1 + \alpha_2 = \theta$; $h^\vee = \langle \rho, \theta \rangle + 1 = 2 + 1 = 3$. The reflection $s_{\varepsilon_i - \varepsilon_j}$ exchanges ε_i and ε_j (e.g. $s_{\alpha_1} \varepsilon_1 = \varepsilon_1 - (\frac{2}{3} + \frac{1}{3})(\varepsilon_1 - \varepsilon_2) = \varepsilon_2$), so $W = S_3$ permuting the ε_i , the symmetry group of an equilateral triangle, D_3 . P^+ is the 60° cone spanned by ω_1, ω_2 ($|\omega_i|^2 = \frac{2}{3}$, $\langle \omega_1, \omega_2 \rangle = \frac{1}{3}$).

P50. $A_1 \times A_1$: $A = \text{Gram} = \text{diag}(2, 2)$. A_2 : both $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. B_2 (α_1 long, α_2 short, $|\alpha_2|^2 = 1$, angle 135°): $\langle \alpha_1, \alpha_2 \rangle = -1$, $\text{Gram} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$, $A = \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$. G_2 (α_2 short, $|\alpha_2|^2 = \frac{2}{3}$, angle 150°): $\langle \alpha_1, \alpha_2 \rangle = -1$, $\text{Gram} \begin{pmatrix} 2 & -1 \\ -1 & 2/3 \end{pmatrix}$, $A = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$. Lattices: in B_2 the short roots α_2 and $\alpha_1 + \alpha_2$ are orthogonal of length 1 ($\langle \alpha_2, \alpha_1 + \alpha_2 \rangle = -1 + 1 = 0$, $|\alpha_1 + \alpha_2|^2 = 2 - 2 + 1 = 1$) and generate Q , since $\alpha_1 = (\alpha_1 + \alpha_2) - \alpha_2$: the unit square lattice; the $A_1 \times A_1$ root lattice is the square lattice of side $\sqrt{2}$. In G_2 the short roots $\pm\alpha_2, \pm(\alpha_1 + \alpha_2), \pm(\alpha_1 + 2\alpha_2)$ have length $\sqrt{2/3}$ and $\langle \alpha_2, \alpha_1 + \alpha_2 \rangle = -1 + \frac{2}{3} = -\frac{1}{3}$, i.e. 120° : they form an A_2 system and generate Q ($\alpha_1 = (\alpha_1 + \alpha_2) - \alpha_2$), a hexagonal lattice, the A_2 lattice scaled by $1/\sqrt{3}$. Weyl groups: reflections in the two long roots of B_2 (which are orthogonal) generate $D_2 = W(A_1 \times A_1)$, index two in D_4 ; the reflection in a short root is the missing generator. The long roots of G_2 , $\pm\alpha_1, \pm(\alpha_1 + 3\alpha_2), \pm(2\alpha_1 + 3\alpha_2)$, form an A_2 system whose reflections generate $D_3 = W(A_2)$, index two in D_6 ; again a short-root reflection is missing.

P51. With α_1, α_2 the sides from the vertex 0 and $\alpha_1 - \alpha_2$ the third side, $\alpha_1 \cdot \alpha_2 = (2+2-6)/2 = -1$, giving A . The squared norm of $m\alpha_1 + n\alpha_2$ is $2m^2 - 2mn + 2n^2 = 2(m^2 - mn + n^2)$; $m^2 - mn + n^2 = \frac{3}{4}n^2 + (m - \frac{n}{2})^2 \geq 1$ for $(m, n) \neq 0$ integers, with equality exactly at $(\pm 1, 0), (0, \pm 1), \pm(1, 1)$: six vectors of norm 2 at mutual angles multiples of 60° , the root system A_2 . Since $|\alpha_i|^2 = 2$, $s_{\alpha_i}(v) = v - (v \cdot \alpha_i)\alpha_i$, and $v \cdot \alpha_i \in \mathbb{Z}$ for lattice v (integral Gram), so each reflection maps Q to itself; $s_1 s_2$ is the rotation by 120° , of order three, and $\langle s_1, s_2 \rangle$ is the dihedral group of order six, S_3 . The dual lattice $\{v : v \cdot Q \subseteq \mathbb{Z}\}$ in root coordinates is $A^{-1}\mathbb{Z}^2$, and $[P : Q] = \det A = 3$. Everything used only the three lengths (the Gram matrix is determined by them) and integrality; no Lie algebra was input. For a boundary triangulation's harmonic Gram to equal A the triangulation would need a harmonic basis whose mass pairing is A , i.e. a D_3 -symmetric metric and marking; a generic triangulation has neither, which is why the reading places the root lattice on an internal fiber (Section 32) rather than on the boundary.

P52. $\mathfrak{su}(2)$: $W = \{1, s\}$, $\varepsilon(s) = -1$, $s(\lambda + \rho) = -(\lambda + 1)\omega$, so with $x = e^\omega$ the numerator is $x^{\lambda+1} - x^{-\lambda-1}$ and the denominator $x - x^{-1}$; the quotient is $x^\lambda(1 - x^{-2(\lambda+1)})/(1 - x^{-2}) = x^\lambda \sum_{i=0}^\lambda x^{-2i}$. $\mathfrak{su}(3)$: $\lambda = a\omega_1 + b\omega_2$, $\rho = \omega_1 + \omega_2$, positive coroots $\alpha_1, \alpha_2, \theta$; $\langle \lambda + \rho, \alpha_1 \rangle = a + 1$, $\langle \lambda + \rho, \alpha_2 \rangle = b + 1$, $\langle \lambda + \rho, \theta \rangle = a + b + 2$, against 1, 1, 2 for ρ : $\dim = (a + 1)(b + 1)(a + b + 2)/2$. Checks: $2 \cdot 1 \cdot 3/2 = 3$, $2 \cdot 2 \cdot 4/2 = 8$, $4 \cdot 1 \cdot 5/2 = 10$, $3 \cdot 3 \cdot 6/2 = 27$.

P53. Proof: $(\sum_w \varepsilon(w)e^{w(\lambda+\rho)})(\sum_\nu m_\mu(\nu)e^\nu) = \sum_w \varepsilon(w) \sum_\nu m_\mu(\nu)e^{w(\lambda+\rho)+\nu}$; substitute $\nu = w\nu'$ (a bijection of P with $m_\mu(w\nu') = m_\mu(\nu')$) to get $\sum_{\nu'} m_\mu(\nu') \sum_w \varepsilon(w)e^{w(\lambda+\rho+\nu')}$. If $\lambda + \rho + \nu'$ is fixed by a reflection s , pairing w with ws cancels the inner sum; otherwise $\lambda + \rho + \nu' = w_{\nu'}^{-1}\kappa$ with κ strictly dominant, and the inner sum is $\varepsilon(w_{\nu'}) \sum_w \varepsilon(w)e^{w\kappa}$, the numerator of $\text{ch } V_{\kappa-\rho}$. Divide by the Weyl denominator. $\mathfrak{su}(2)$, $V_1 \otimes V_\lambda$: $\nu = \pm 1$ gives $V_{\lambda+1}$ and, for $\lambda \geq 1$, $V_{\lambda-1}$; at $\lambda = 0$, $\lambda + \nu + \rho = 0$ is on the wall and is killed: $V_1 \otimes V_0 = V_1$. $V_2 \otimes V_0$: weights of V_2 are 2, 0, -2; $\nu = 2$ gives V_2 , $\nu = 0$ gives V_0 , $\nu = -2$ gives $\lambda + \nu + \rho = -1$, folded by s to $+1 = \rho + 0$ with sign -1: $-V_0$. Total $V_2 + V_0 - V_0 = V_2$. $\mathfrak{su}(3)$, $\mathbf{3} \otimes \mathbf{3}$ ($\lambda = (1, 0)$, weights of $\mathbf{3}$: $(1, 0)$, $(-1, 1)$, $(0, -1)$ in the ω basis): $(2, 0) = \mathbf{6}$; $(0, 1) = \mathbf{\bar{3}}$; $(1, -1) + \rho = (2, 0)$ lies on the wall $\langle \cdot, \alpha_2^\vee \rangle = 0$ and is killed: $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \mathbf{\bar{3}}$ ($9 = 6 + 3$). $\mathbf{3} \otimes \mathbf{\bar{3}}$ ($\lambda = (0, 1)$): $(1, 1) = \mathbf{8}$; $(-1, 2) + \rho = (0, 3)$ on a wall, killed; $(0, 0) = \mathbf{1}$: $\mathbf{8} \oplus \mathbf{1}$.

P54. A_1 : $e^\omega - e^{-\omega} = e^\omega(1 - e^{-2\omega}) = e^\rho(1 - e^{-\alpha})$. A_2 : $e^\rho(1 - e^{-\alpha_1})(1 - e^{-\alpha_2})(1 - e^{-\alpha_1 - \alpha_2})$ expands to eight terms, $e^\rho[1 - e^{-\alpha_1} - e^{-\alpha_2} - e^{-\alpha_1 - \alpha_2} + e^{-\alpha_1 - \alpha_2} + e^{-2\alpha_1 - \alpha_2} + e^{-\alpha_1 - 2\alpha_2} - e^{-2\alpha_1 - 2\alpha_2}]$; the two $e^{-\alpha_1 - \alpha_2}$ terms cancel, leaving six. They are the W -orbit of ρ with signs: ρ (1, +); $s_1\rho = \rho - \alpha_1$ (-); $s_2\rho = \rho - \alpha_2$ (-); $s_1s_2\rho = \rho - \alpha_1 - s_1\alpha_2 = \rho - 2\alpha_1 - \alpha_2$ (+); $s_2s_1\rho = \rho - \alpha_1 - 2\alpha_2$ (+); $s_1s_2s_1\rho = -\rho = \rho - 2\alpha_1 - 2\alpha_2$ (-).

Affine algebras and theta functions (Section 20)

P55. $wt_\gamma w^{-1}(\mu) = w(w^{-1}\mu + K\gamma) = \mu + Kw\gamma = t_{w\gamma}(\mu)$, so $t(Q^\vee)$ is normalized by W (W preserves $Q^\vee$); a translation fixes no point unless $\gamma = 0$ while W fixes 0, so $W \cap t(Q^\vee) = 1$ and the group generated is $W \ltimes t(Q^\vee)$. Since θ is long, $\theta^\vee = \theta$ and $s_0(\mu) = s_\theta(\mu) + K\theta = \mu - (\langle \mu, \theta \rangle - K)\theta$, the reflection in the affine hyperplane $\langle \mu, \theta \rangle = K$ along θ : it fixes that hyperplane pointwise and squares to the identity. $\mathfrak{su}(2)$, $K = k + 2$, $\theta = \alpha = 2\omega$: $s_0(m\omega) = -m\omega + 2K\omega = (2(k + 2) - m)\omega$, fixing $m = k + 2$. Alcove: the dominant chamber is a simplicial cone with r walls, and $\theta^\vee = \sum_i a_i^\vee \alpha_i^\vee$ with all $a_i^\vee > 0$, so the half-space $\langle \mu, \theta^\vee \rangle \leq K$ cuts it to a bounded simplex. Interior lattice points satisfy $\langle \mu, \alpha_i^\vee \rangle \geq 1$ and $\langle \mu, \theta^\vee \rangle \leq K - 1$; with $\mu = \lambda + \rho$ these read $\langle \lambda, \alpha_i^\vee \rangle \geq 0$ and $\langle \lambda, \theta^\vee \rangle \leq K - 1 - \langle \rho, \theta^\vee \rangle = K - h^\vee = k$: exactly $P_k^+ + \rho$.

P56. $e^{t_\gamma(\hat{\mu})} = e^{\mu + K\gamma} e^{-\langle \mu, \gamma \rangle + \frac{K}{2}|\gamma|^2} = e^{\mu + K\gamma} q^{\langle \mu, \gamma \rangle + \frac{K}{2}|\gamma|^2}$, and $\frac{K}{2}|\frac{\mu}{K} + \gamma|^2 = \frac{|\mu|^2}{2K} + \langle \mu, \gamma \rangle + \frac{K}{2}|\gamma|^2$, so $q^{\langle \mu, \gamma \rangle + \frac{K}{2}|\gamma|^2} = q^{-|\mu|^2/2K} q^{\frac{K}{2}|\frac{\mu}{K} + \gamma|^2}$; with $e^{\mu + K\gamma} = e^{K(\frac{\mu}{K} + \gamma)} \mapsto e^{2\pi i K \langle \frac{\mu}{K} + \gamma, z \rangle}$ the sum over γ is $q^{-|\mu|^2/2K} \Theta_\mu^{(K)}$. Every $\hat{w} \in \hat{W}$ is uniquely $t_\gamma w$; $\varepsilon(t_\gamma) = +1$ because $t_{\theta^\vee} = s_0 s_\theta$ is a product of two reflections and the $t_{w\theta^\vee}$ generate $t(Q^\vee)$. Hence $\sum_{\hat{w}} \varepsilon(\hat{w}) e^{\hat{w}(\hat{\lambda} + \hat{\rho})} = \sum_{w \in W} \varepsilon(w) \sum_\gamma e^{t_\gamma(w(\hat{\lambda} + \hat{\rho}))} = q^{-|\lambda + \rho|^2/2K} \sum_w \varepsilon(w) \Theta_{w(\lambda + \rho)}^{(K)}$, the prefactor being common because W preserves norms. The lattice enters through $Q^\vee$ and the quadratic form $|\cdot|^2$ (the theta function); the non-abelian structure enters only through W and its sign, and through ρ and $h^\vee$ in the shift $\lambda \mapsto \lambda + \rho$, $k \mapsto k + h^\vee$.

P57. $h_\lambda - \frac{c}{24} = \frac{|\lambda + \rho|^2 - |\rho|^2}{2(k + h^\vee)} - \frac{k \dim \mathfrak{g}}{24(k + h^\vee)}$. The strange formula gives $|\rho|^2 = h^\vee \dim \mathfrak{g}/12$, so $\frac{|\rho|^2}{2(k + h^\vee)} + \frac{k \dim \mathfrak{g}}{24(k + h^\vee)} = \frac{(h^\vee + k) \dim \mathfrak{g}}{24(k + h^\vee)} = \frac{\dim \mathfrak{g}}{24}$. $\mathfrak{su}(2)$: $\dim \mathfrak{g} = 3$, $|\rho|^2 = |\omega|^2 = \frac{1}{2}$, $h^\vee = 2$, and

$\frac{1}{2}/4 = \frac{1}{8} = \frac{3}{24}$ checks the strange formula; $h_\lambda = (\frac{(\lambda+1)^2}{2} - \frac{1}{2})/2(k+2) = \lambda(\lambda+2)/4(k+2)$, $T_\lambda = \exp 2\pi i[(\lambda+1)^2/4(k+2) - \frac{1}{8}]$. Central charges: $\mathfrak{su}(2)_1$: $3/3 = 1$; $\mathfrak{su}(2)_2$: $6/4 = \frac{3}{2}$; $\mathfrak{su}(3)_1$: $8/4 = 2$. The $\mathfrak{su}(2)_1$ doublet: $h_0 = 0$, $h_1 = 3/12 = \frac{1}{4}$, so $T = e^{-2\pi i/24} \text{diag}(1, e^{2\pi i/4}) = e^{-\pi i/12} \text{diag}(1, i)$, which is $\rho_2(T) = \text{diag}(e^{\pi i j^2/2}) = \text{diag}(1, i)$ up to the overall phase.

P58. $Q = \sqrt{2}\mathbb{Z}$, $\omega = 1/\sqrt{2}$, so $Q + j\omega = \{\sqrt{2}(n + \frac{j}{2})\}$ and $\Theta_{Q+j\omega} = \sum_n q^{(n+j/2)^2} e^{2\pi i \sqrt{2}(n+j/2)z}$, which is $\theta_j^{(2)}(z/\sqrt{2}; \tau) = \sum_n \exp(2\pi i(n + \frac{j}{2})^2 \tau + 4\pi i(n + \frac{j}{2})z/\sqrt{2})$. So $\chi_j = \theta_j^{(2)}/\eta$. Under $\tau \mapsto \tau + 1$: $\theta_j^{(2)} \mapsto e^{\pi i j^2/2} \theta_j^{(2)}$ and $\eta \mapsto e^{\pi i/12} \eta$, so $T = e^{-\pi i/12} \text{diag}(1, i)$, agreeing with P57. Under $\tau \mapsto -1/\tau$: by P31, $\theta_j(z/\tau; -1/\tau) = (-i\tau)^{1/2} e^{\pi i k z^2/\tau} \cdot k^{-1/2} \sum_l e^{-2\pi i j l/k} \theta_l(z; \tau)$ with $k = 2$, and $\eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau)$; the $(-i\tau)^{1/2}$ cancels, leaving $S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. These are the Kac–Peterson matrices of P64 at $k = 1$. Fusion: Verlinde with $S = H$ gives $N_{11}^0 = \sum_\sigma S_{1\sigma}^2 = 1$ and $N_{11}^1 = \sum_\sigma S_{1\sigma}^3/S_{0\sigma} = 0$: $1 \times 1 = 0$, addition in $\mathbb{Z}/2$, which is the composition $\mathbf{X}^a \mathbf{X}^b = \mathbf{X}^{a+b}$ of characteristics in the Heisenberg–Weyl group of P23.

$\mathfrak{su}(3)_1$: the cosets are Q , $Q + \omega_1$, $Q + \omega_2$ with $\omega_2 \equiv 2\omega_1$; Poisson summation over the even lattice Q (dual P) gives $\Theta_{Q+a}(-1/\tau, z/\tau) = (-i\tau)^{(\dots)} |P/Q|^{-1/2} \sum_b e^{-2\pi i \langle a, b \rangle} \Theta_{Q+b}(\tau, z)$, and η^2 contributes $(-i\tau)$, so $S_{ab} = 3^{-1/2} e^{-2\pi i \langle \omega_a, \omega_b \rangle}$. With $\langle \omega_1, \omega_1 \rangle = \langle \omega_2, \omega_2 \rangle = \frac{2}{3}$, $\langle \omega_1, \omega_2 \rangle = \frac{1}{3}$: $S_{11} = S_{22} = e^{2\pi i/3}/\sqrt{3}$, $S_{12} = e^{4\pi i/3}/\sqrt{3}$, i.e. $S_{ab} = 3^{-1/2} e^{2\pi i ab/3}$ for $a, b \in \mathbb{Z}/3$, the Fourier transform of $\mathbb{Z}/3$. T : $h_a = |\omega_a|^2/2 = \frac{1}{3}$ for $a = 1, 2$ and $c = 2$, so $T = e^{-2\pi i/12} \text{diag}(1, e^{2\pi i/3}, e^{2\pi i/3})$. Parity: on $Q(A_2)$, $|n\alpha_1 + m\alpha_2|^2 = 2n^2 - 2nm + 2m^2$ is even, so $q^{|\gamma|^2/2}$ is invariant under $\tau \mapsto \tau + 1$ term by term and each coset picks up the constant $e^{\pi i |\omega_a|^2}$ (since $|\omega_a + \gamma|^2 = |\omega_a|^2 + 2\langle \omega_a, \gamma \rangle + |\gamma|^2$ with $\langle \omega_a, \gamma \rangle \in \mathbb{Z}$). The rank-one register at level 3 is the lattice $\sqrt{3}\mathbb{Z}$ with $|\gamma|^2 = 3n^2$, odd for odd n : $\tau \mapsto \tau + 1$ multiplies the terms by $(-1)^n$, which depends on n , and $\theta_j^{(3)}$ is not an eigenfunction — the obstruction of P30. Evenness of the lattice, not the rank or the level, is what makes T diagonal.

Rank two, the alcove, fusion and density (Sections 21–25)

P59. Averaging any positive form over a finite $G \subset \text{GL}(2, \mathbb{Z})$ gives a G -invariant one, so G is a finite subgroup of $O(2)$: cyclic C_n or dihedral D_n . An element of $\text{GL}(2, \mathbb{Z})$ has integer trace, and a rotation by $2\pi/n$ has trace $2\cos(2\pi/n) \in \mathbb{Z}$ only for $n \in \{1, 2, 3, 4, 6\}$. Up to conjugacy: C_1, C_2, C_3, C_4, C_6 and D_1, D_2, D_3, D_4, D_6 (some with two classes). Reflections: D_2 is generated by reflections in two orthogonal lines — the Weyl group of $A_1 \times A_1$; D_3 by reflections at 60° in the hexagonal lattice — $W(A_2)$; D_4 by reflections in the axes and diagonals of the square lattice — $W(B_2)$; D_6 by the six reflections of the hexagonal lattice — $W(G_2)$.

P60. Grid with the $(i, j) - (i+1, j+1)$ diagonal: the edge set is preserved by I , the swap $(i, j) \mapsto (j, i)$, negation $(i, j) \mapsto (-i, -j)$, and their product; the 90° rotation sends the diagonal to the antidiagonal, which is not an edge. Group $\{\pm I, \pm \sigma\} \cong D_2$, abelian. The triangular lattice torus, with edges in three directions at 60° , is preserved by every element of D_6 : non-abelian. An automorphism of a finite complex permutes finitely many simplices, so it has finite order, whereas $T_B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has infinite order in $\text{GL}(2, \mathbb{Z})$. A flip pass replaces the far triangulation by a different one (the grid in a sheared basis) and stacks one tetrahedron per flipped edge, so the near and far faces are different triangulations of the same torus; the far marking is read through the lattice map, giving $M = L^\top$, and the quarter turn is the diagonal pass followed by the relabelling. This is a change of complex, not a symmetry of one complex.

P61. $\mathfrak{su}(2)$: $\theta^\vee$ is the coroot, $\langle \lambda\omega, \alpha^\vee \rangle = \lambda$, so $\lambda \in \{0, \dots, k\}$: $k+1$ states. $\mathfrak{su}(3)$: $\langle \lambda, \theta^\vee \rangle = \lambda_1 + \lambda_2 \leq k$: $\binom{k+2}{2} = (k+1)(k+2)/2$ states; three at $k=1$ and six at $k=2$.

P62. Units: fundamental weight ω with $|\omega|^2 = \frac{1}{2}$, root $\alpha = 2\omega$, $|\alpha|^2 = 2$, coroot $\alpha^\vee = \alpha$, $Q^\vee = \mathbb{Z}\alpha$, $\rho = \omega$, $h^\vee = 2$. For $\mu = m\omega$ and $\gamma = n\alpha = 2n\omega$, $\mu/K + \gamma = (m/K + 2n)\omega$ and $|\mu/K + \gamma|^2 = \frac{1}{2}(2n + m/K)^2 = 2(n + \frac{m}{2K})^2$, so $\Theta_m^{(K)} = \sum_n \exp(2\pi i K(n + \frac{m}{2K})^2 \tau + 2\pi i K(2n + \frac{m}{K}) \langle \omega, z \rangle)$, which is the level- $2K$ theta function of characteristic $m \bmod 2K$ in the variable $w = \langle \omega, z \rangle = \frac{1}{2} \langle \alpha, z \rangle$. $W = \{\pm 1\}$ acts by $m \mapsto -m$ with sign -1 , so $A_m^{(K)} = \theta_m^{(2K)} - \theta_{-m}^{(2K)}$. The numerator has $\mu = \lambda + \rho = (\lambda+1)\omega$ at level $K = k + h^\vee = k+2$, giving $\theta_{\lambda+1}^{(2(k+2))} - \theta_{-(\lambda+1)}^{(2(k+2))}$; the denominator has $\mu = \rho = \omega$ at level $h^\vee = 2$, giving $\theta_1^{(4)} - \theta_{-1}^{(4)}$. Numerically, at $\tau = 0.13 + i$ and $w = 0.17 + 0.06i$, both sides agree to 7×10^{-16} for $\lambda = 0, 1$; and for $k = 1, 2, 3$ the quotient divided by $q^{h_\lambda - c/24}$ tends to the finite character $\sum_{m \equiv \lambda(2), |m| \leq \lambda} e^{2\pi i m w}$ as $\text{Im } \tau \rightarrow \infty$. At $w = 0$ the numerator and the denominator both vanish, because $\theta_{-m}(0) = \theta_m(0)$ by evenness ($w = 0$ lies on the Weyl wall); the value there is a removable limit, which is why the reading requires evaluation away from the walls or a controlled limit.

P63. With no positive roots the Weyl denominator is the empty product, 1; W trivial makes $A_\mu = \Theta_\mu$; $\rho = 0$ and $h^\vee = 0$ leave $\chi_\lambda = \Theta_\lambda^{(k)}$, the level- k theta function of the charge $\lambda \in \mathbb{Z}/k$: Definition 10.1 of the reading. In the non-abelian ratio the denominator is $A_\rho^{(h^\vee)}$, the alternating sum at level $h^\vee$; substituting $h^\vee = 0$ there gives a theta function at level zero, which is not defined (the Gaussian factor disappears and the lattice sum diverges). The abelian case is a separate construction, not a limit of the ratio.

P64. Symmetry is visible. Unitarity: with $N = k+2$ and $m = \mu+1 \in \{1, \dots, N-1\}$, $\frac{2}{N} \sum_{m=1}^{N-1} \sin \frac{\pi a m}{N} \sin \frac{\pi b m}{N} = \delta_{ab}$ for $1 \leq a, b \leq N-1$ (discrete sine transform), so $SS^\top = I$; S is real symmetric unitary, hence $S^2 = I$, the (trivial) charge conjugation of $\mathfrak{su}(2)$. $k=1$: $S = \sqrt{\frac{2}{3}} \cdot \frac{\sqrt{3}}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, Hadamard; $T = \text{diag}(e^{-\pi i/12}, e^{5\pi i/12}) = e^{-\pi i/12} \text{diag}(1, i)$. $k=2$: $S = \begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{pmatrix}$ (the Ising S -matrix), $T = \text{diag}(e^{-\pi i/8}, e^{\pi i/4}, e^{7\pi i/8})$. Exactness of $(ST)^3 = S^2$: at $k=1$, $ST = e^{-\pi i/12} H \text{diag}(1, i)$ and P32 gives $(H \text{diag}(1, i))^3 = e^{\pi i/4} I$, so $(ST)^3 = e^{-3\pi i/12} e^{\pi i/4} I = I = S^2$; the $c/24$ phase $e^{-2\pi i c/24}$ with $c=1$ exactly cancels the metaplectic phase. The same holds at every k (numerically to 6×10^{-16} for $k \leq 4$): the modular data of a rational theory is a true representation of $\text{SL}(2, \mathbb{Z})$ once T carries $e^{-2\pi i c/24}$.

P65. $k=2$, labels 0, 1, 2. $N_{11}^0 = \sum_\sigma S_{1\sigma}^2 = 1$; $N_{11}^1 = \sum_\sigma S_{1\sigma}^2 S_{1\sigma}/S_{0\sigma} = \frac{1}{2} \cdot \frac{1}{\sqrt{2}}/\frac{1}{2} + 0 + \frac{1}{2} \cdot (-\frac{1}{\sqrt{2}})/\frac{1}{2} = 0$; $N_{11}^2 = \frac{1}{2} \cdot \frac{1}{2}/\frac{1}{2} + 0 + \frac{1}{2} \cdot \frac{1}{2}/\frac{1}{2} = 1$. So $1 \times 1 = 0 + 2$ (Ising: $\sigma \times \sigma = 1 + \psi$), matching the rule $|\lambda - \mu| \leq \nu \leq \min(\lambda + \mu, 2k - \lambda - \mu)$, $\lambda + \mu + \nu$ even. Kac-Walton: the weights of V_1 are ± 1 ; $1+1=2$ and $1-1=0$ both lie in the alcove, so $N_{11}^0 = N_{11}^2 = 1$ with no reflection. For 2×1 : $2+1=3$ lies on the affine wall $\lambda+1 = k+2$ and is fixed by the affine reflection, contributing zero, while $2-1=1$ gives $N_{21}^1 = 1$ ($\psi \times \sigma = \sigma$). Against Proposition 19.2 (P53) the only new ingredient is the affine reflection s_0 in the wall $\lambda+1 = k+2$; the finite reflection s acts as before.

P66. With $x = e^{i\phi}$, $\text{ch } V_\lambda(x) = (x^{\lambda+1} - x^{-\lambda-1})/(x - x^{-1}) = \sin((\lambda+1)\phi)/\sin \phi$, and $\phi = -\pi(\sigma+1)/(k+2)$ gives the ratio of sines (the sign of ϕ cancels). At $k=2$, $K=4$: $v_1 = (\sqrt{2}, 0, -\sqrt{2})$, $v_2 = (1, -1, 1)$, $v_0 = (1, 1, 1)$ for $\sigma = 0, 1, 2$; then $v_1^2 = (2, 0, 2) = v_0 + v_2$, the Ising rule $1 \times 1 = 0 + 2$

evaluated pointwise. The evaluation matrix $E_{\lambda\sigma} = S_{\lambda\sigma}/S_{0\sigma}$ is S with its columns rescaled by $1/S_{0\sigma} \neq 0$, hence invertible since S is unitary. By Verlinde, $\sum_{\nu} N_{\lambda\mu}^{\nu} S_{\nu\sigma} = S_{\lambda\sigma} S_{\mu\sigma}/S_{0\sigma}$ (multiply the formula by $S_{\nu\sigma'}$ and sum over ν using unitarity), so $\sum_{\nu} N_{\lambda\mu}^{\nu} v_{\nu}(\sigma) = v_{\lambda}(\sigma) v_{\mu}(\sigma)$: fusion is pointwise multiplication of the vectors v_{λ} , and the structure constants are recovered by changing back with E^{-1} .

P67. (a) $Q^{\vee} = Q = \mathbb{Z}^r$ in root coordinates (simply laced), and $P = A^{-1}\mathbb{Z}^r$, so $|P/KQ^{\vee}| = |A^{-1}\mathbb{Z}^r/K\mathbb{Z}^r| = |\mathbb{Z}^r/KA\mathbb{Z}^r| = K^r \det A$: $2K$ for A_1 and $3K^2$ for A_2 . (b) With $c = A^{-1}m_{\mu}/K$ (m_{μ} the Dynkin labels), $\Theta_{\mu}(\tau, \xi) = \sum_{n \in \mathbb{Z}^r} f(n + c)$, $f(x) = e^{\pi i K \tau x^{\top} A x + 2\pi i K x^{\top} A \xi}$. Poisson summation gives $\sum_n f(n + c) = \sum_{m \in \mathbb{Z}^r} \hat{f}(m) e^{2\pi i m \cdot c}$ with

$$\begin{aligned} \hat{f}(m) &= \int e^{\pi i K \tau x^{\top} A x + 2\pi i x^{\top} (K A \xi - m)} dx \\ &= (-iK\tau)^{-r/2} (\det A)^{-1/2} \exp\left(-\frac{\pi i}{K\tau} (K A \xi - m)^{\top} A^{-1} (K A \xi - m)\right). \end{aligned}$$

Expanding, the exponent is $-\pi i K \xi^{\top} A \xi / \tau + 2\pi i m^{\top} \xi / \tau - \pi i m^{\top} A^{-1} m / (K\tau)$, and with $c' = A^{-1}m/K$ the last two terms are $2\pi i K c'^{\top} A (\xi / \tau) + \pi i K (-1/\tau) c'^{\top} A c'$: the summand of $\Theta_{\mu'}(-1/\tau, \xi/\tau)$ for the characteristic m . Since $e^{2\pi i m \cdot c} = e^{2\pi i m^{\top} A^{-1} m_{\mu}/K}$ is constant on cosets $m + KA\mathbb{Z}^r$, grouping gives

$$\Theta_{\mu}(\tau, \xi) = (-iK\tau)^{-r/2} (\det A)^{-1/2} e^{-\pi i K \xi^{\top} A \xi / \tau} \sum_{\mu'} e^{2\pi i \langle \mu, \mu' \rangle / K} \Theta_{\mu'}(-1/\tau, \xi/\tau).$$

Replacing (τ, ξ) by $(-1/\tau, \xi/\tau)$, using $\Theta_{\mu'}(\tau, -\xi) = \Theta_{-\mu'}(\tau, \xi)$ and relabelling $\mu' \rightarrow -\mu'$ gives the stated formula, with $(iK/\tau)^{-r/2} (\det A)^{-1/2} = (-i\tau)^{r/2} (K^r \det A)^{-1/2}$. For the character ratio, numerator (level $k + h^{\vee}$) and denominator (level $h^{\vee}$) share $(-i\tau)^{r/2}$, and their exponentials give $e^{\pi i (k + h^{\vee} - h^{\vee}) \xi^{\top} A \xi / \tau} = e^{\pi i k \xi^{\top} A \xi / \tau}$; the constants $|P/KQ^{\vee}|^{-1/2}$ are absorbed in the normalization of S . Numerically, for A_2 at $K = 2$ the identity holds to 3×10^{-15} over the 12 characteristics. (c) The lattice $\mathbb{Z}^r + \tau A \mathbb{Z}^r$ with the theta functions above carries the polarization KA (the quasi-periodicity factor under $\xi \mapsto \xi + \tau e_j$ is $e^{-\pi i K \tau A_{jj} - 2\pi i K (A\xi)_j}$), whose section count is $\det(KA) = K^r \det A$, not the K^r of a principal polarization; so τA is a period matrix for a nonprincipal bundle. The genus- r Siegel inversion $\Omega \mapsto -\Omega^{-1}$ would send τA to $-A^{-1}/\tau$, a different lattice, whereas the transformation derived above stays on $\Omega = (-1/\tau)A$ and mixes the $|P/KQ^{\vee}|$ cosets.

P68. Ng–Schauenburg: the torus representation factors through $\mathrm{SL}(2, \mathbb{Z}/N)$, a finite group, for every modular tensor category, so no level and no algebra gives infinitely many operators on the torus. Freedman–Larsen–Wang gives density only for the braid group of a surface with enough labelled punctures; universality therefore needs (i) punctures carrying alcove labels, (ii) braid generators exchanging them. In the code, a puncture is a removed top cell (the hole register: holes as removed top cells, holonomy read around them), its label the sector of the harmonic space it carries, and a braid generator a surgical move that exchanges two holes' positions; the conformal block is the harmonic space of the punctured host at the register's degree. None of these has been built as a labelled puncture with a fusion space yet; the reading lists them as optional native constructions.

P69. Hypothesis: the level- k register of a boundary torus M for a compact simple $\mathfrak{g}$ is the theta space of the lattice $H^1(M; \mathbb{Z}) \otimes Q^{\vee}$ — genus- r theta functions with period matrix $\tau_M \cdot \mathrm{Gram}(Q^{\vee})$, the nonprincipal polarization of P67 and characteristics in $P/KQ^{\vee}$, $K = k + h^{\vee}$ — antisymmetrized under W and divided by A_{ρ} ; the surface supplies only $H^1(M; \mathbb{Z})$ and τ_M , the algebra supplies $Q^{\vee}$, W and $(\rho, h^{\vee})$, and the mapping class group of M acts through S, T on τ_M by the Kac–Peterson

matrices. The Weyl group acts on the fiber, not on the triangulation: a D_6 -symmetric torus realizes D_6 as mapping classes of M (elements of $\mathrm{GL}(2, \mathbb{Z})$ acting on H^1), which coincides with $W(G_2)$ as a group but is not the W of the register, so its symmetry buys nothing. First experiment (synthetic, the objective V5): at τ fixed, build the antisymmetrized theta functions for $\mathfrak{su}(3)_k$ (and $\mathfrak{su}(2)_k$ as the genus-one control), fit S and T from the transformation law with the level- k automorphy factor, and compare with the Kac–Peterson closed forms and the Verlinde fusion; then take τ from a seeded host and the flip-realized S, T of the torus. Falsified if the fit residual is not at machine precision, if the fitted matrices fail the closed forms, or if the fusion coefficients are not the integers of Kac–Walton. The geometric realization of the fiber would be a genus- r boundary component on the locus $\Omega = \tau \cdot \mathrm{Gram}(Q^\vee)$.

The experiment specification

Whole harmonic states, pairings and elimination (Sections 26–28)

P70. Necessity: if $B = TA$ then $\ker A \subseteq \ker B$, and “defined on every input” means every input is Ac for some c , i.e. A is onto. Sufficiency: let C be a right inverse, $AC = I$. For any c , $A(c - CAc) = 0$, so $c - CAc \in \ker A \subseteq \ker B$ and $Bc = BCAC$; hence $T := BC$ satisfies $B = TA$. Two right inverses differ by a map D with $AD = 0$, whose image lies in $\ker A \subseteq \ker B$, so $BD = 0$ and T is the same. Uniqueness: T is determined on $\mathrm{im} A$, which is everything. The equivalent test is $B(I - CA) = 0$, since $I - CA$ projects onto $\ker A$. Examples: $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ (rank one) is not onto: the input e_2 cannot be prescribed (incomplete coverage). With $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, $e_2 \in \ker A$ but $Be_2 = e_2 \neq 0$: a nonzero output for zero input, so T is ambiguous. The incomplete-coverage and ambiguous-output fixtures of NA2 exercise these, together with the hidden-mode fixture (an invisible class in $\ker A \cap \ker B$, which is allowed) and the zero-map fixture.

P71. With $\alpha = d_U f$: $\alpha(v_1 v_2) = g_{12} f(v_2) - f(v_1)$, $\alpha(v_0 v_2) = g_{02} f(v_2) - f(v_0)$, $\alpha(v_0 v_1) = g_{01} f(v_1) - f(v_0)$, so $(d_U \alpha)(v_0 v_1 v_2) = g_{01} g_{12} f(v_2) - g_{01} f(v_1) - g_{02} f(v_2) + f(v_0) + g_{01} f(v_1) - f(v_0) = (g_{01} g_{12} - g_{02}) f(v_2)$. This vanishes for all f iff $g_{01} g_{12} g_{02}^{-1} = 1$, the holonomy around the triangle. A gauge transformation $f(v) \mapsto \lambda_v f(v)$, $g_{ij} \mapsto \lambda_i g_{ij} \lambda_j^{-1}$ preserves d_U ; a chain coefficient $c(v_i v_j)$ must transform by λ_i^{-1} (the inverse system) for $\sum \alpha(e) c(e)$ to be invariant, so the adjoint of d_U is the boundary operator twisted by U^{-1} : $\langle d_U \alpha, c \rangle = \langle \alpha, \partial^{U^{-1}} c \rangle$. A cochain image $Z = G^U \Phi$ is closed iff it pairs to zero with every U^{-1} -boundary, $(\partial_2^{U^{-1}})^T Z = 0$; a chain frame Φ is coclosed (a harmonic chain) iff its U -boundary vanishes, $\partial_1^U \Phi = 0$, which is the adjoint statement $\delta_U \omega = 0$ in chain coordinates. If U is not flat, $d_U^2 \neq 0$ and there is no cohomology to certify, which is why cohomological gluing needs a new justification there.

P72. A boundary class with period vector p is represented by the chain Φc with $Pc = p$, i.e. $c = P^{-1}p$; a dual class with periods $p^\vee$ by $\Phi^\vee (P^\vee)^{-1} p^\vee$. Their bilinear pairing is

$$(\Phi^\vee (P^\vee)^{-1} p^\vee)^T M_{\mathrm{own}} \Phi P^{-1} p = (p^\vee)^T (P^\vee)^{-T} B_0 P^{-1} p,$$

so $G_{\mathrm{per}} = (P^\vee)^{-T} B_0 P^{-1}$, and dropping B_0 is legitimate only when the frames are biorthonormal, $B_0 = I$. Transport sends $p_A \mapsto p_B = M p_A$ and $p_A^\vee \mapsto M^\vee p_A^\vee$; the pairing is preserved iff $(M^\vee p^\vee)^T G_B (M p) = (p^\vee)^T G_A p$ for all $p, p^\vee$, i.e. $(M^\vee)^T G_B M = G_A$. Under a gauge with primal

and dual factors $\rho, \tilde{\rho}$ the dressed mass transforms as $G' = \tilde{\rho}^{-\top} G \rho^{-1}$, so $(\tilde{\rho}\tilde{c})^\top G'(\rho c) = \tilde{c}^\top G c$: invariant. A primal–primal Gram transforms to $c^\top \rho^\top \tilde{\rho}^{-\top} G \rho^{-1} \rho c$, which is not $c^\top G c$ unless $\tilde{\rho} = \rho$, i.e. at zero connection (P15).

P73. The second block row of $Kx = 0$ is $K_{zb}b + K_{zz}z = 0$, so $z = -K_{zz}^{-1}K_{zb}b$; substituting in the first row gives $(K_{bb} - K_{bz}K_{zz}^{-1}K_{zb})b = 0$. Every solution is recovered from b by the formula for z , so nothing is discarded: the internal coordinates are determined, not summed over. For the quadratic $E(u) = \|u\|^2 + \|r - T_2u\|^2$, $\nabla_{\bar{u}}E = u - T_2^\dagger(r - T_2u) = 0$ gives $(I + T_2^\dagger T_2)u = T_2^\dagger r$. Then $r - T_2u = (I - T_2(I + T_2^\dagger T_2)^{-1}T_2^\dagger)r = (I + T_2T_2^\dagger)^{-1}r$ (push-through identity), and $E = r^\dagger[T_2(I + T_2^\dagger T_2)^{-2}T_2^\dagger + (I + T_2T_2^\dagger)^{-2}]r$; writing $T_2(I + T_2^\dagger T_2)^{-2}T_2^\dagger = (I + T_2T_2^\dagger)^{-2}T_2T_2^\dagger$ the bracket is $(I + T_2T_2^\dagger)^{-2}(T_2T_2^\dagger + I) = (I + T_2T_2^\dagger)^{-1}$. The zero-energy set is $r = 0$, i.e. $z = T_2T_1x$: composition. A partial trace would replace the internal block by a mixed marginal, losing the correlations between b and z ; the Schur complement keeps z as a function of b , so the joint solution is reconstructible.

Quantum norms, channels and acceptance (Sections 29–30)

P74. $\hat{T}^\dagger \hat{T} = Q_{\text{in}}^{-1/2} T^\dagger Q_{\text{out}} T Q_{\text{in}}^{-1/2}$, which is I iff $T^\dagger Q_{\text{out}} T = Q_{\text{in}}$. With $|\hat{T}\rangle\rangle = \sum_{ij} \hat{T}_{ij} |i\rangle_{\text{out}} |j\rangle_{\text{ref}}$, $\text{Tr}_{\text{out}} |\hat{T}\rangle\rangle \langle\langle \hat{T}| = \sum_{jj'} (\sum_i \hat{T}_{ij} \overline{\hat{T}_{ij'}}) |j\rangle \langle j'|$, and $\sum_i \overline{\hat{T}_{ij'}} \hat{T}_{ij} = (\hat{T}^\dagger \hat{T})_{j'j}$, so the marginal is $(\hat{T}^\dagger \hat{T})^\top$ divided by the norm $\|\hat{T}\|_F^2 = \text{Tr} \hat{T}^\dagger \hat{T}$; for an isometry this is I/d_{in} . The Schmidt decomposition of a vector in $\mathbb{C}^{d_{\text{out}}} \otimes \mathbb{C}^{d_{\text{in}}}$ is the singular value decomposition of its coefficient matrix, so the Schmidt coefficients are $\sigma_i / \|\hat{T}\|_F$.

P75. Each $|K_a\rangle\rangle \langle\langle K_a|$ is positive semidefinite, hence so is the sum; conversely a positive J has a spectral decomposition into such terms, which is the Choi criterion for complete positivity. By P74, $\text{Tr}_{\text{out}} |K_a\rangle\rangle \langle\langle K_a| = (K_a^\dagger K_a)^\top$, so $\text{Tr}_{\text{out}} J = (\sum_a K_a^\dagger K_a)^\top$, which is I iff the Kraus operators are trace preserving. $V^\dagger V = \sum_{ab} K_a^\dagger K_b \langle a|b\rangle = \sum_a K_a^\dagger K_a = I$, and $\text{Tr}_E(V\rho V^\dagger) = \sum_{ab} K_a \rho K_b^\dagger \text{Tr}(|a\rangle \langle b|) = \sum_a K_a \rho K_a^\dagger$. Dimension count: $\dim(B \oplus E) = d_B + d_E$ while $\dim(B \otimes E) = d_B d_E$; a relation into a direct sum assigns to each input a pair (component in B , component in E), and “discarding” E is a projection onto a summand, which is linear on vectors, whereas the partial trace over a tensor factor is a map on operators that produces mixed states. No linear projection of a direct sum reproduces Tr_E , so a third boundary is not an environment in the Stinespring sense.

P76. (a) $H^2(T^2 \times T^2) = H^0 \otimes H^2 \oplus H^1 \otimes H^1 \oplus H^2 \otimes H^0$ has dimension $1 + 4 + 1 = 6$; the two degree-one registers occupy the multidegree-(1,1) summand $H^1 \otimes H^1$, of dimension 4, not all of H^2 . (b) k^g ; for two tori, $k \cdot k = k^2$, the tensor product $\Theta_k \otimes \Theta_k$ (Proposition 11.1). (c) $V_1 \otimes V_1 = V_0 \oplus V_2$ has dimension 4 and two summands; at level 1 the alcove is $\{0, 1\}$ and the fusion is $1 \times 1 = 0$ only, the summand V_2 being truncated by the affine wall (P65). (d) A two-qubit gate has $4 \times 4 = 16$ amplitudes and its operator vector lives in $\Theta_2 \otimes \Theta_2 \otimes \overline{\Theta_2 \otimes \Theta_2}$, also 16; a direct sum of two two-dimensional port frames has $2 + 2 = 4$ amplitudes and its operators are block 2×2 maps with $4 + 4 = 8$ entries; a selected-pair χ is one 2×2 matrix, 4 entries. The state space $\Theta_2 \otimes \Theta_2$ also has 4 amplitudes, but a tensor product is more than a count: it comes with a factorization of the basis and of the operator algebra ($D_{(p_1, p_2), (q_1, q_2)} = D_{p_1, q_1} \otimes D_{p_2, q_2}$), and a direct sum has neither.

P77. Oracle comparisons: R1–R4 (the read M against the predicted φ^* : identity or swap), C1 (the group law $\sigma^2 = I$), D1 (the predicted direct sum), T4 (fitted $\rho_2(S), \rho_2(T)$ against the closed forms $H, \text{diag}(1, i)$), T5 and T6 (fitted matrices against the Kronecker product and against

controlled- Z), T7 (fits at two moduli against each other), T8 (the geometric monodromy's fit against the generator formulas), F2 (measured passes against T_A, T_B, S), F3 (the far modulus against the bridge formula), F4 and T9 (the two orders against the matrix products), G2 (the tube monodromy against the block form), E2 (the decagon monodromy against the prediction from the side classes), E4–E5 (unitarity, order and the entangling witness against Clifford enumeration), E6 (the product-collar control). Certificates: R5 (isotropy for the declared form, with the flipped control as a negative), H1–H3 (closedness, coclosedness, independent span), J1–J3 (metric independence and exactness of the difference), U1 (the transport defect reported, not predicted), T1–T3 (fit residual, unitarity, projective group law on held-out samples), F1 (the flipped host is a manifold with the declared boundary), G1 and G3 (Betti numbers, isotropy, symmetry and positivity of the read period matrix), G4–G5 (entropy of the geometric state and the degeneration trend, reported), E1 and E3 (Euler characteristic, automorphism, fixed period matrix).

Neck geometry and encoded-state entropy (Section 33)

P78. (a) The binary entropy is $h(p) = -p \log_2 p - (1-p) \log_2(1-p)$ and $-(1-p) \log_2(1-p) = (1-p)(p + p^2/2 + \dots)/\ln 2 = p/\ln 2 + O(p^2)$, so $h(p) = -p \log_2 p + p/\ln 2 + O(p^2)$. With $p = C\epsilon^2 + O(\epsilon^3)$, $-p \log_2 p = -C\epsilon^2 \log_2(C\epsilon^2) + O(\epsilon^3 \log \epsilon)$, giving the stated expansion; the log term dominates $C\epsilon^2$, so S/ϵ^2 diverges as $\epsilon \rightarrow 0$. (b) The exponent contains $\pi i k(n + j/k)^\top \Omega(n + j/k)$ whose ϵ -derivative is $2\pi i k(n_1 + j_1/k)(n_2 + j_2/k)$; hence $\partial_\epsilon \theta_j|_0 = \sum_{n_1, n_2} 2\pi i k m_1 m_2 e^{(\dots)_1} e^{(\dots)_2} = \frac{1}{2\pi i k} \left(\sum_{n_1} 2\pi i k m_1 e^{(\dots)_1} \right) \left(\sum_{n_2} 2\pi i k m_2 e^{(\dots)_2} \right) = \frac{\theta'_{j_1}(z_1) \theta'_{j_2}(z_2)}{2\pi i k}$, since $\theta'_j(z) = \sum_n 2\pi i k(n + j/k) e^{(\dots)}$. At level two, $\theta_0(-z) = \theta_0(z)$ (substitute $n \mapsto -n$) and $\theta_1(-z) = \theta_1(z)$ (substitute $n \mapsto -n - 1: -(n + \frac{1}{2}) \mapsto n + \frac{1}{2}$), so $\theta'_j(0) = 0$ and the first-order change of the value column at $z = 0$ vanishes. The normalized coherent state therefore has no $O(\epsilon)$ component normal to the product manifold; the leading entangling component is $O(\epsilon^2)$, the small singular value is $O(\epsilon^2)$ and $p \sim C\epsilon^4$. (c) Build $\Omega(\epsilon)$, evaluate `coherent_state([0,0], Omega)` and `entanglement_entropy(vector, (2,2))` at $\epsilon = 0.04, 0.02, 0.01$; the smaller probability falls by 16 per halving (measured slope 3.999 on a log-log plot, values $4.5 \times 10^{-7}, 2.8 \times 10^{-8}, 1.8 \times 10^{-9}$), while a random product state perturbed to first order gives slope 2 and the remainder $(S - [-p \log_2 p + p/\ln 2])/\epsilon^2 \rightarrow 0$.

The measured constructions

Reading the records (Section 35)

P79. Expected counts: 17/17 for each two-torus fixture, 27/27 for the four-torus fixtures; theta 8/8 (two tori) and 10/10 (four tori). The map: R1–R4, rank and integrality of M against φ^* : Theorem 4.2 and $M = \varphi^*$ (Section 4); R5, isotropy for Ω_∂ with the flipped control: Corollary 3.2 and Remark 3.3; H1–H3: the certificates that replace Theorem 2.2 on complex masses; J1–J3: Theorem 2.4 (P3); C1: Theorem 6.1's group law (P14); D1: Proposition 16.1; U1: Remark 7.1 and Proposition 7.2; T1: Theorem 13.2 (closure of the theta space); T2: Proposition 13.4; T3: the projective group law; T4: the S and T formulas of Section 13; T5: Propositions 11.1 and 13.3; T6: the shear/controlled- Z formula; T7: Ω -independence of ρ_k ; T8: the read monodromy acting on the register. With an arbitrary closed two-dimensional band the periods would still be periods of classes, so R1–R4, R5, C1, D1 and the theta checks (which use only M) would pass; H2 (coclosedness)

and H3 (independent kernel of the same span) would fail, and J2–J3 would change meaning (the representative would not be selected by the metric). This is why the harmonic identification is certified separately from the topological reads.

P80. (a) L_b maps $(1, 0) \mapsto (1, 1)$, $(0, 1) \mapsto (0, 1)$, $(1, 1) \mapsto (1, 2)$; the flip removes the row edges $(1, 0)$ and adds the edges $(0, -1) - (1, 1)$ of displacement $(1, 2)$, so the flipped edge set $\{(1, 1), (0, 1), (1, 2)\}$ is exactly L_b of the grid's edge set: the flipped grid is the grid in the sheared basis, relabelled by $(x, y) \mapsto (x, y+x)$. A cycle with new-coordinate homology class v has old class $L_b v$, so the far periods are read through L_b and the collar's relation is $M = L_b^\top = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = T_B$. Likewise $L_a^\top = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = T_A$ and $L_s^\top = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = S$; with a collar twist the passes compose as $M = L_{\text{last}}^\top \cdots L_{\text{first}}^\top \cdot M_{\text{twist}}$. (b) The second **b** pass flips the new-coordinate row edges, whose old displacement is $L_b(1, 0)^\top = (1, 1)^\top$, and creates edges of old displacement $L_b(1, 2)^\top = (1, 3)^\top$. On the 3-periodic grid $(1, 3) \equiv (1, 0)$: the new edge joins the same vertex pair as an original row edge, which is still an edge of the host (it is the bottom of a first-pass tetrahedron). Two edges on one vertex pair are not simplicial, so the code refuses the pass by name. On the 5×5 grid $(1, 3) \not\equiv (1, 0)$, and two passes fit. (c) Expected: 20/20 for one pass on grid 3, 21/21 for two passes on grid 5; F2 reads T_B, T_A, S to 4×10^{-15} ; F3 gives $\tau' = (M_{21} + M_{22}\tau)/(M_{11} + M_{12}\tau)$ to 8×10^{-16} ; F4 reads $M = L_b^\top L_a^\top = T_B T_A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ for **ab** (pass **a** first, then **b**; the last pass acts on the left) and $T_A T_B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ for **ba**, which differ.

P81. (a) Attaching a tube (an annulus $S^1 \times [0, 1]$, $\chi = 0$) to two tori after removing a disk from each ($\chi = -1$ each) gives $\chi = -1 - 1 + 0 = -2$, a connected closed orientable surface of genus two. $\chi(\partial W) = \chi(T^2) + \chi(T^2) + \chi(\Sigma_2) = -2$, so $\chi(W) = -1$ (a compact odd-dimensional manifold has half the Euler characteristic of its boundary). By P44 the tube is a 1-handle joining two collars, so $H^1(W) = H^1(W_1) \oplus H^1(W_2) = \mathbb{Z}^4$ and $b_1 = 4$; then $1 - 4 + b_2 - b_3 = -1$ with $b_3 = 0$ (nonempty boundary) gives $b_2 = 2$: $[1, 4, 2, 0]$. $b_1(\partial W) = 2 + 2 + 4 = 8$ and the restriction has rank 4 (Corollary 3.2). (b) The relation is a property of $H^1(W)$, which splits by the 1-handle argument: each harmonic class restricts to periods on one collar's tori only, so $M = M_1 \oplus M_2$ (measured I_4 or $\text{swap} \oplus \text{swap}$, off-block norm 2.6×10^{-15}). The period matrix of Σ_2 is defined by its own harmonic forms in the boundary metric, whose supports cross the neck; Ω_{12} measures that leakage and is a metric quantity of one connected surface, not of the relation. (c) Measured on the default neck: $\text{Im } \Omega$ positive with eigenvalues 0.83, 1.08, $\Omega_{12} = 1.45 \times 10^{-4} - 1.88 \times 10^{-4}i$, symmetric to 2.4×10^{-15} ; entropy 0 for the diagonal control and 4×10^{-7} bits mean; over the 25 necks $|\Omega_{12}|$ runs from 5×10^{-8} to 2.4×10^{-2} and falls approximately exponentially with tube length, while the diagonal approaches the two tori's moduli as the neck pinches. In the plumbing description a genus-two surface near the separating degeneration is two tori joined by an annulus of modulus ℓ/w (length over circumference), and its period matrix is $\text{diag}(\tau_1, \tau_2)$ plus corrections linear in the plumbing parameter t , with $\Omega_{12} \propto t$ and $|t| \sim e^{-2\pi\ell/w}$ (Fay's degeneration formula): exponentially small in the neck's length-to-waist ratio, as observed.

P82. The decagon with opposite sides identified ($E_{i+5} \sim E_i^{-1}$) is a genus-two surface; coning the ten sides to a centre and barycentrically subdividing gives 28 vertices (3 identified corner/centre points, 5 side midpoints, 10 spoke midpoints, 10 barycentres), 90 edges and 60 faces, $28 - 90 + 60 = -2$. The rotation permutes the vertices and preserves the face list, so `is_automorphism` returns true; it has order ten as a simplicial isometry of the flat decagon metric. `induced_map(2)` returns M on (A_1, A_2, B_1, B_2) with $\det M = 1$, $M^\top J M = J$ and $M^5 = I$ (P47), mixing the two factors. The `fit` `fit = ThetaRegister(level=2).weil(M, omega)` returns `fit.matrix` unitary with residual $\sim 6 \times 10^{-16}$ and $\rho^5 \propto I$ to 1.5×10^{-15} ; `ThetaRegister.schmidt_rank(fit.matrix, (2,2))` is 4. Applying `fit.apply` to the 36 products of X, Y, Z eigenstates and calling `entanglement_entropy(v,`

$(2,2)$) gives exactly one bit for 16 of them and zero for the rest; the gate is at projective distance 1.00 from every $A \otimes B$ and 0.77 from every $\text{SWAP} \cdot A \otimes B$ over the 24×24 single-qubit Cliffords (E5). The CLI run reports 13/13; r^4 gives order five and Schmidt rank two, $r^5 = -I$ Schmidt rank one and no entangling power, and the product collar reads $M = I$ with entropy 5×10^{-29} (E6).

P83. (a) With the flat metric of modulus τ the cotangent weights are those of the flat torus; the harmonic cochains are the constant forms dx , dy , and the holomorphic combination $dx + \tau dy$ (P20 and Section 26) has A -period 1 and B -period τ , so $\Omega = \tau$; the cup pairing $C_{ab} = \sum_{\text{faces}} \text{area}(W_a \times W_b)$ evaluated on the harmonic basis gives the intersection matrix $-PC^{-1}P^\top = J$ with $A \cdot B = +1$, the sign the code calibrates on this case. (b) `s.periods.omega` is symmetric to machine precision with $\text{Im } \Omega$ positive definite (eigenvalues 0.46, 1.66); with P, Q, R, S the blocks of `s.induced_map(2)`, $(R + S\Omega)(P + Q\Omega)^{-1}$ equals Ω to 1.5×10^{-15} (E3). A simplicial isometry φ of the surface preserves the metric, hence the Hodge star and the space of holomorphic forms; it acts on $H^1(\Sigma; \mathbb{Z})$ by M_{per} and on the normalized holomorphic basis by the change of marking of P36, so the period matrix in the transported marking is $\Omega' = \Omega$: Ω is a fixed point of M on the Siegel upper half-space, which is why the read matrix is fixed by M . Being fixed by an order-five element singles out a special point of $\mathcal{H}_2$ (the curve $y^2 = x^5 - 1$), but nothing forces it onto the locus $\Omega = \tau A$ with A a root Gram matrix; that locus is a different one-parameter family, and the root register also needs the nonprincipal polarization KA (P67), which a principally polarized Jacobian does not carry.